

BFS Traversal

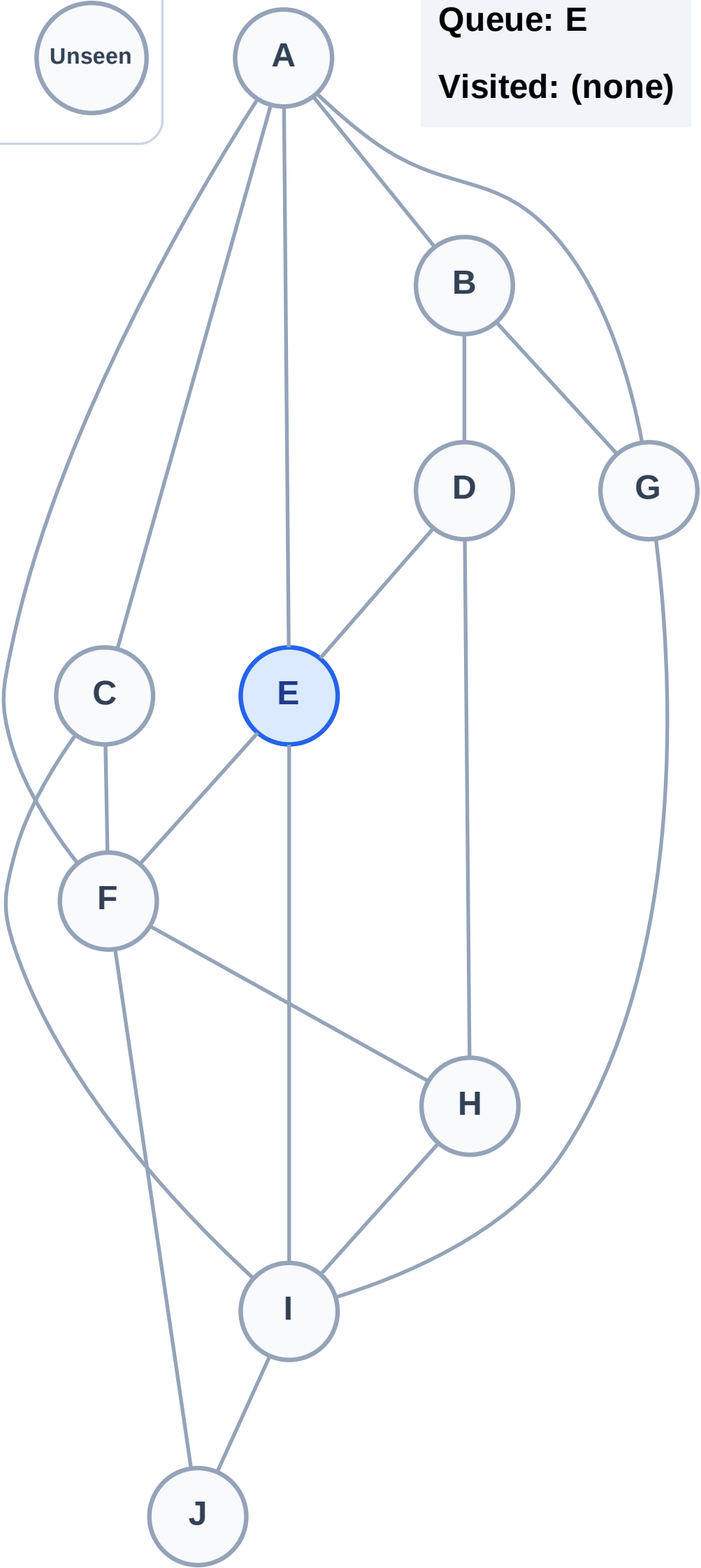
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

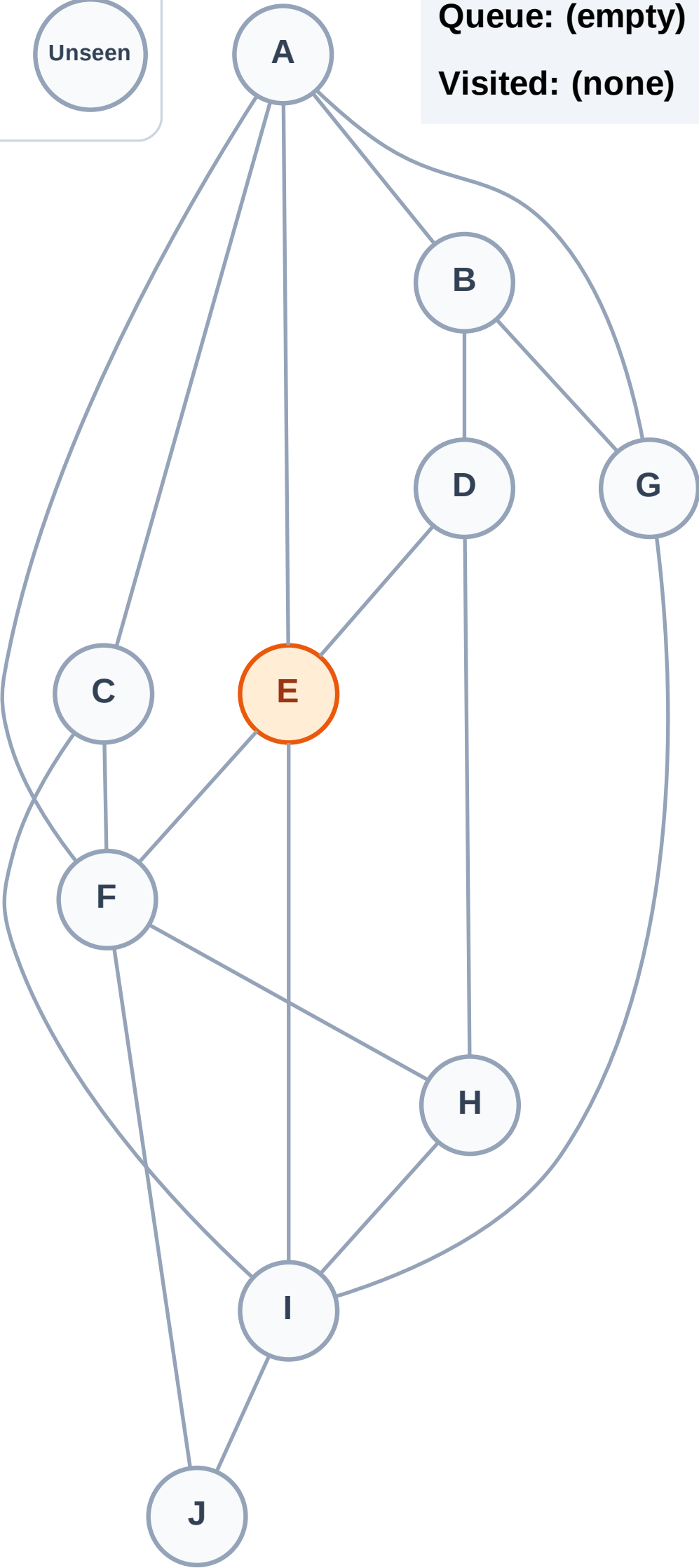
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

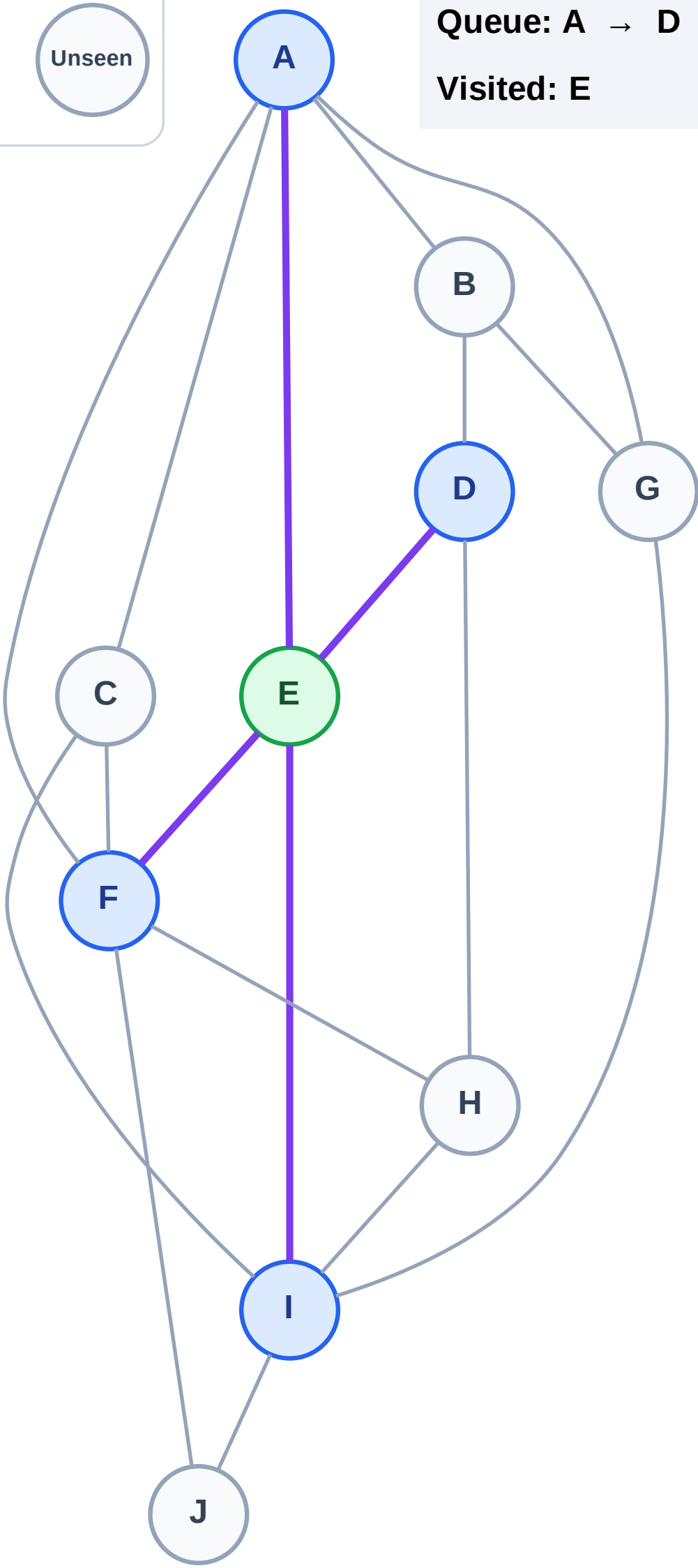
Step 3: Discover A, D, F, I from E

Legend



Queue: A → D → F → I

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

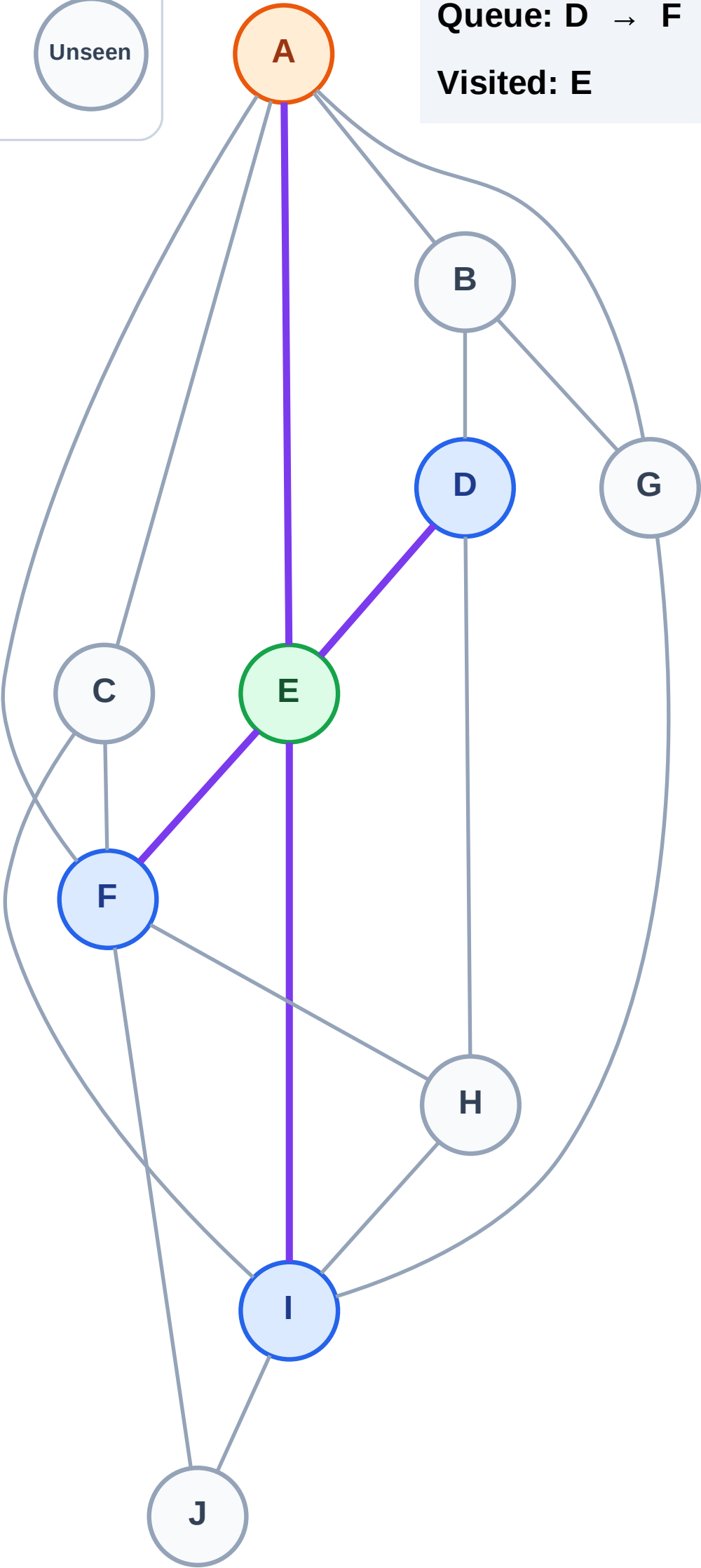
Processing

Discovered

Unseen

Queue: D → F → I

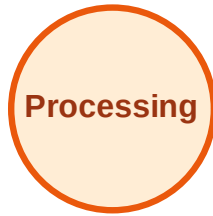
Visited: E



BFS Traversal

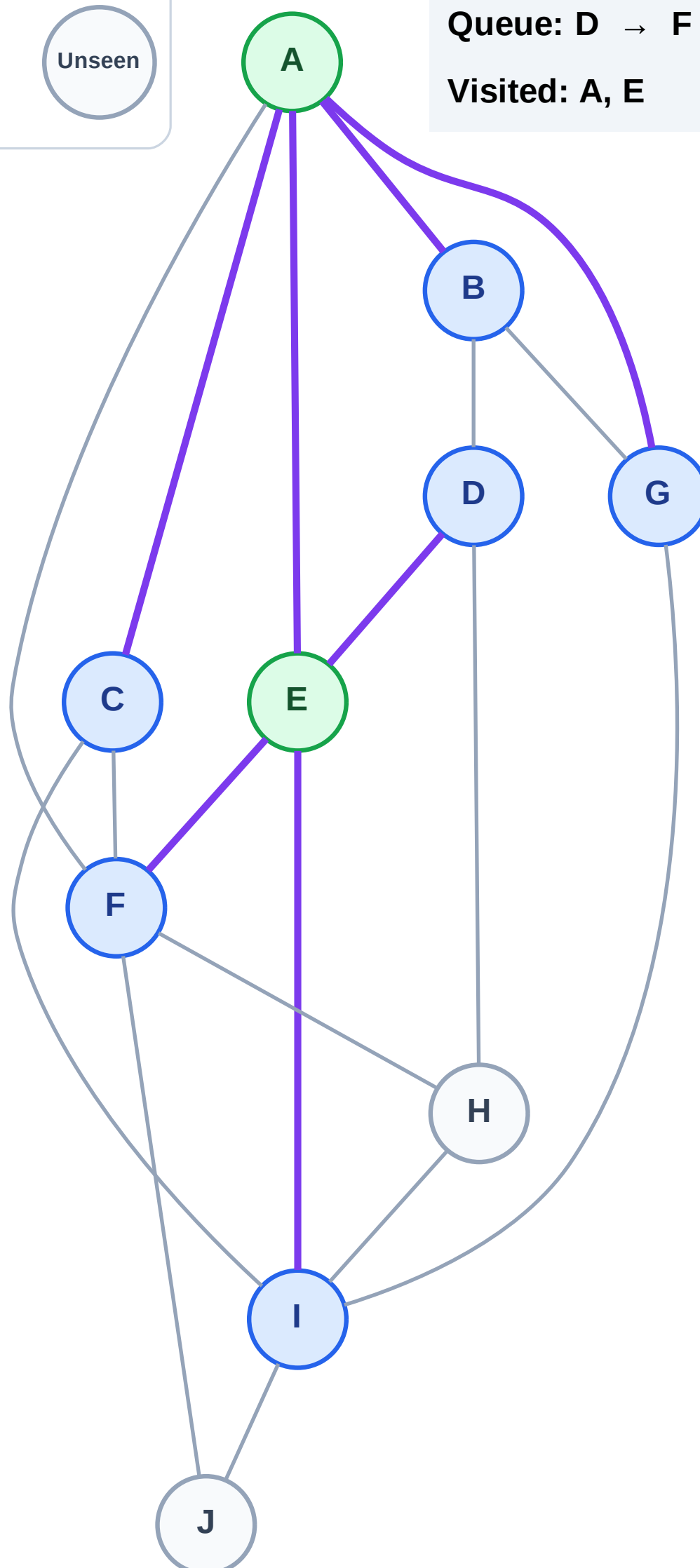
Step 5: Discover B, C, G from A

Legend



Queue: D → F → I → B → C → G

Visited: A, E



BFS Traversal

Step 6: Process node D

Legend

Complete

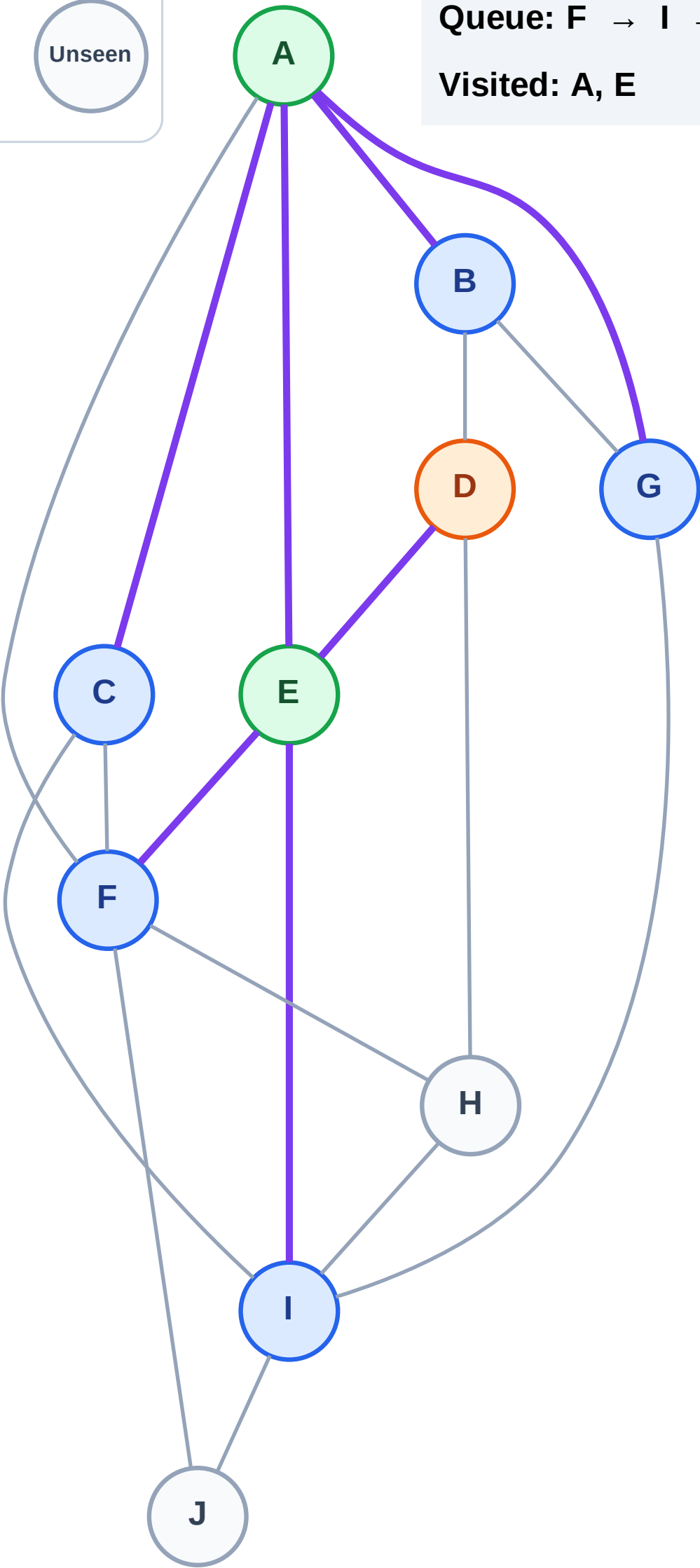
Processing

Discovered

Unseen

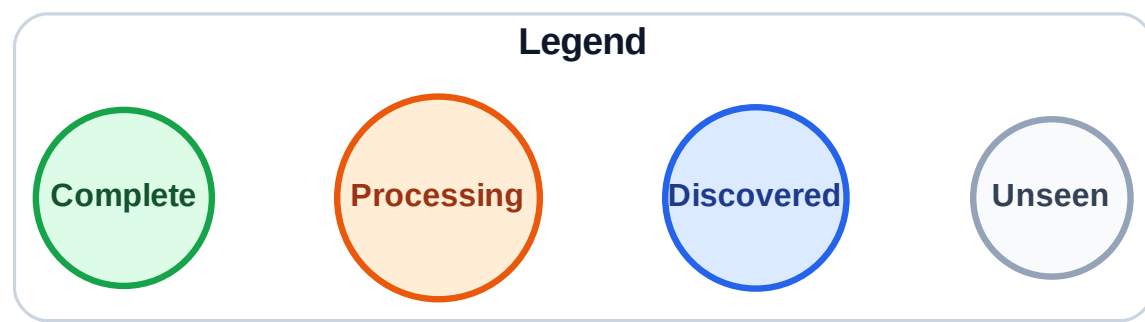
Queue: F → I → B → C → G

Visited: A, E



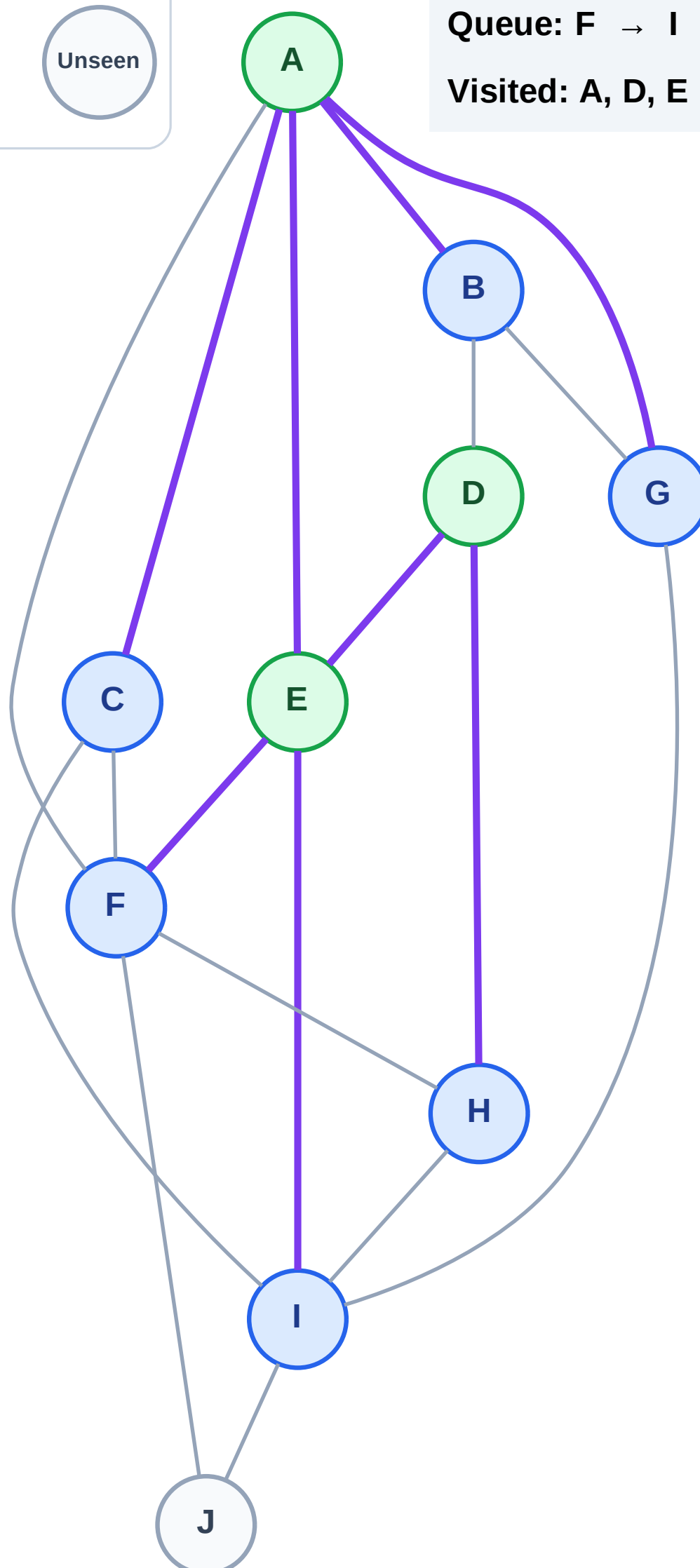
BFS Traversal

Step 7: Discover H from D



Queue: F → I → B → C → G → H

Visited: A, D, E



BFS Traversal

Step 8: Process node F

Legend

Complete

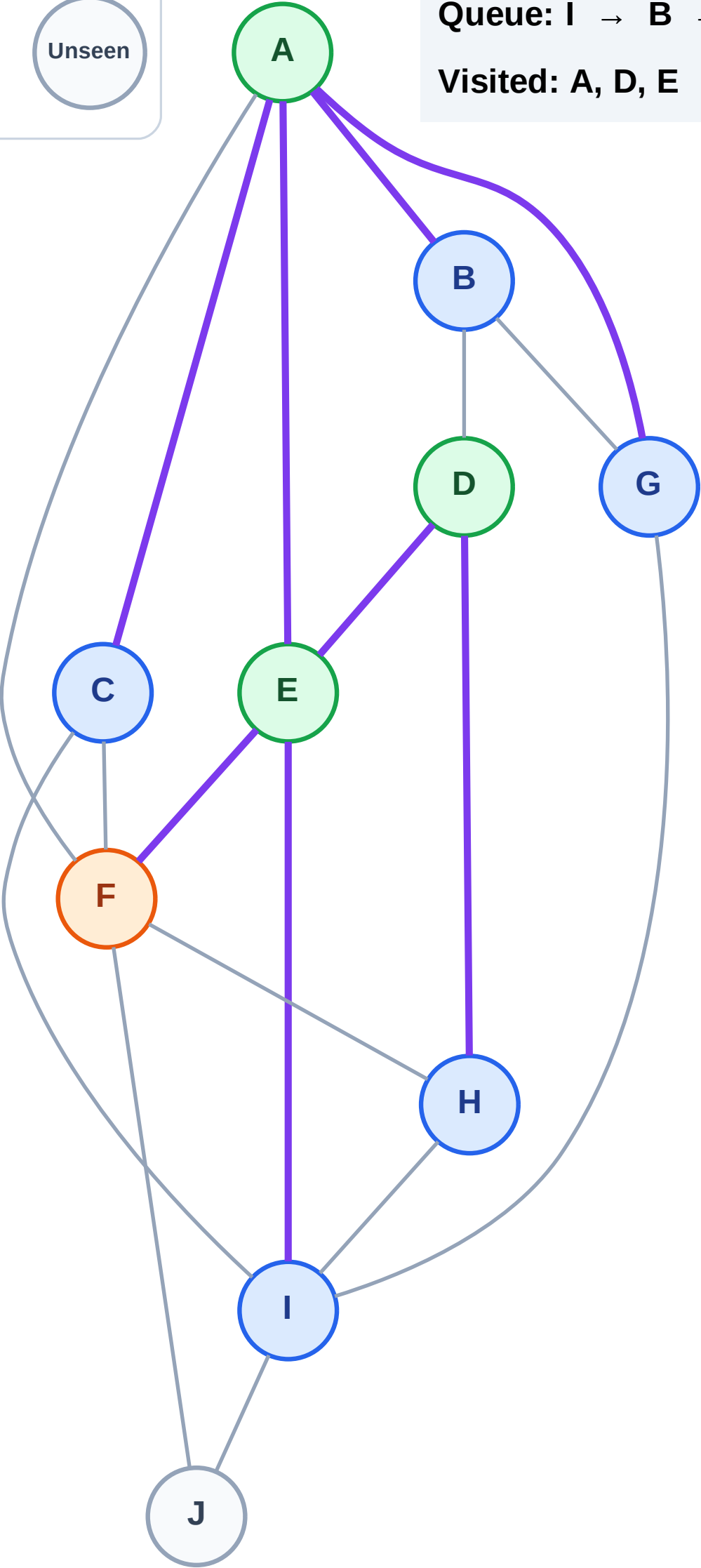
Processing

Discovered

Unseen

Queue: I → B → C → G → H

Visited: A, D, E



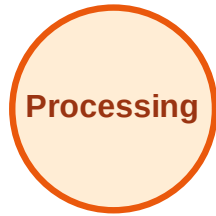
BFS Traversal

Step 9: Discover J from F

Legend



Complete



Processing



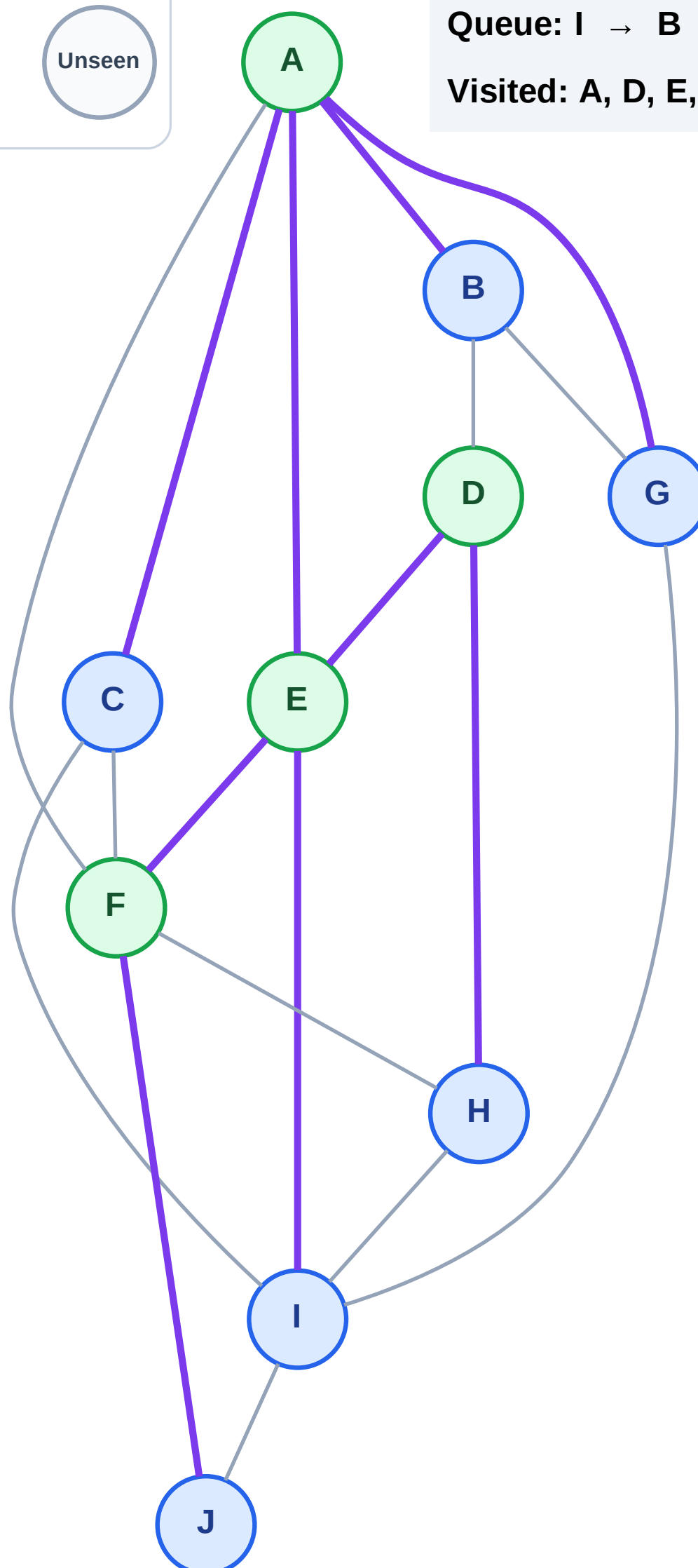
Discovered



Unseen

Queue: I → B → C → G → H → J

Visited: A, D, E, F



BFS Traversal

Step 10: Process node I

Legend

Complete

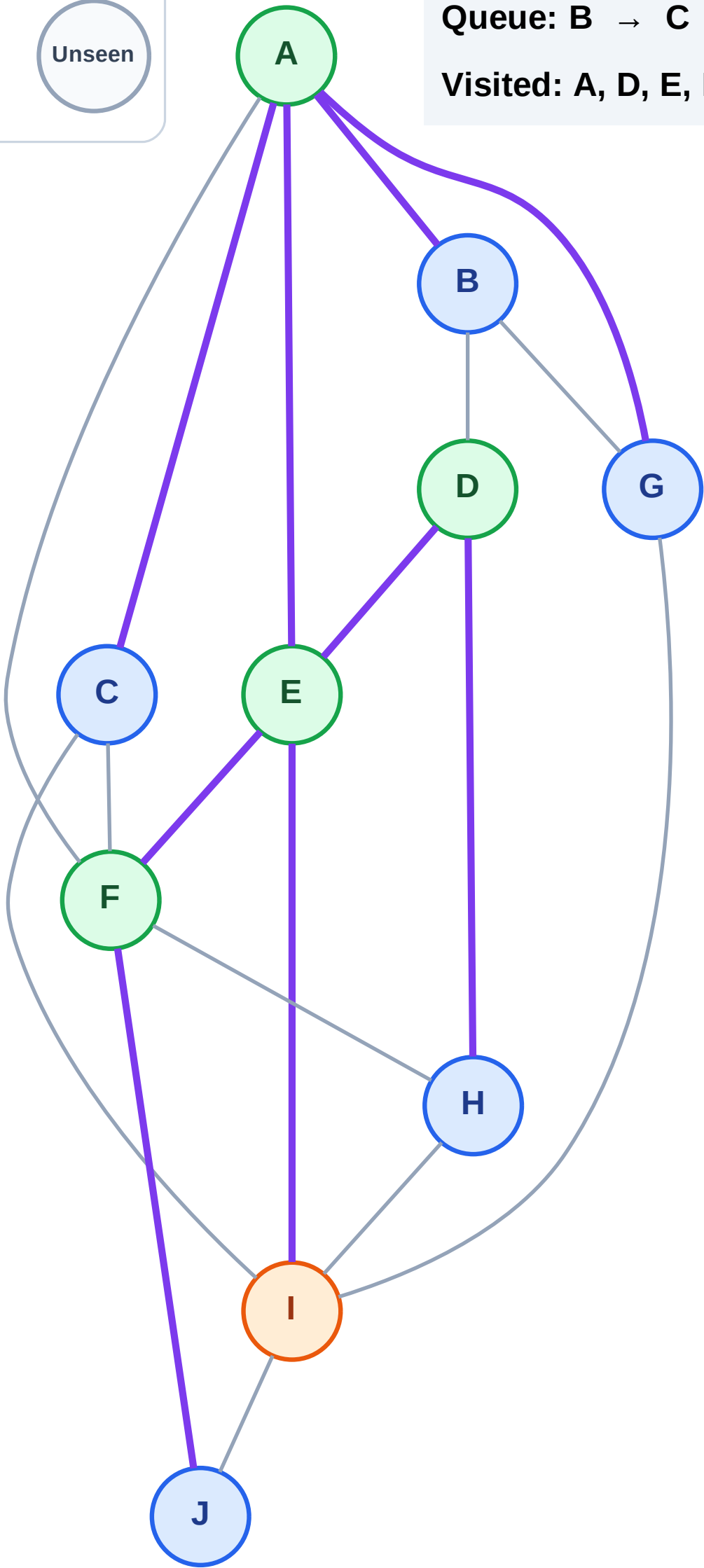
Processing

Discovered

Unseen

Queue: B → C → G → H → J

Visited: A, D, E, F



BFS Traversal

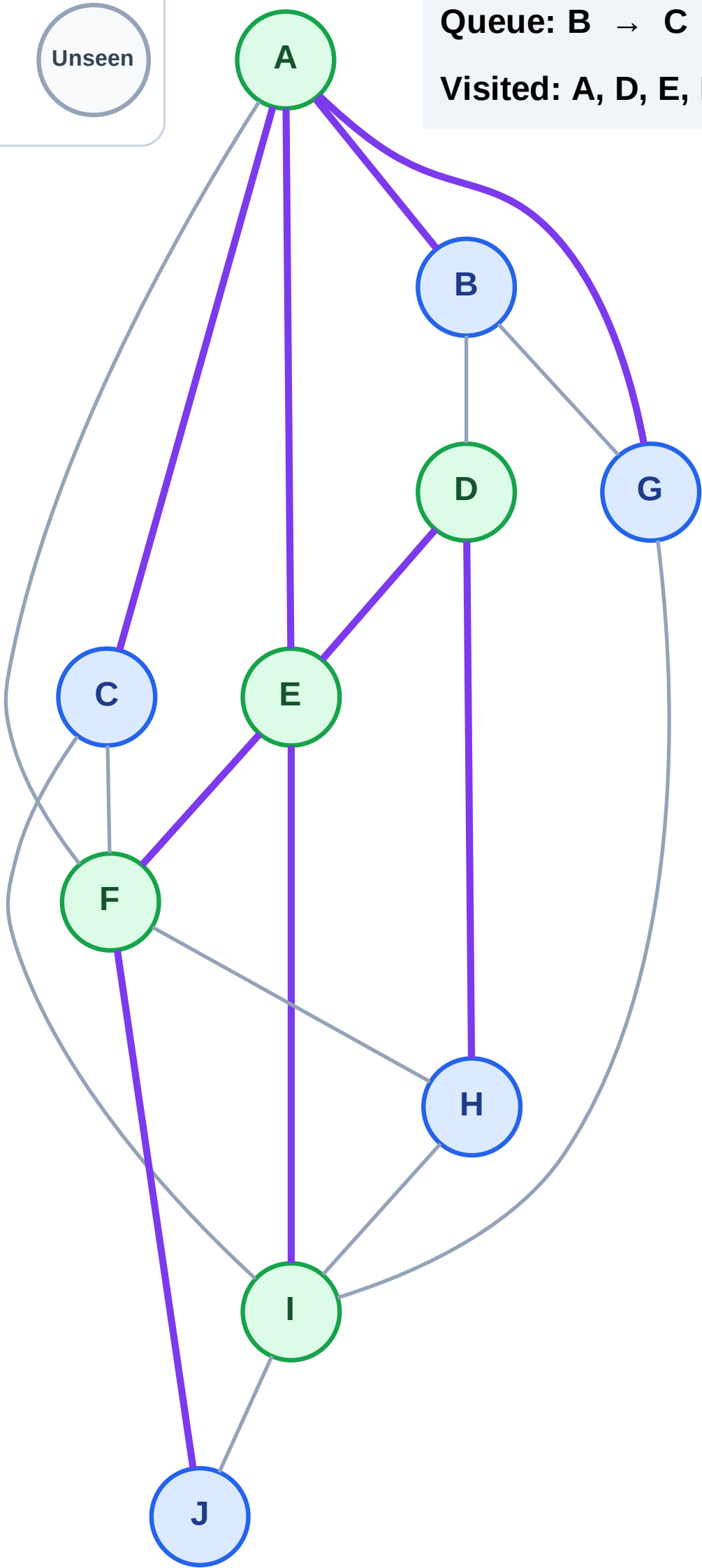
Step 11: No new neighbors from I

Legend



Queue: B → C → G → H → J

Visited: A, D, E, F, I



BFS Traversal

Step 12: Process node B

Legend

Complete

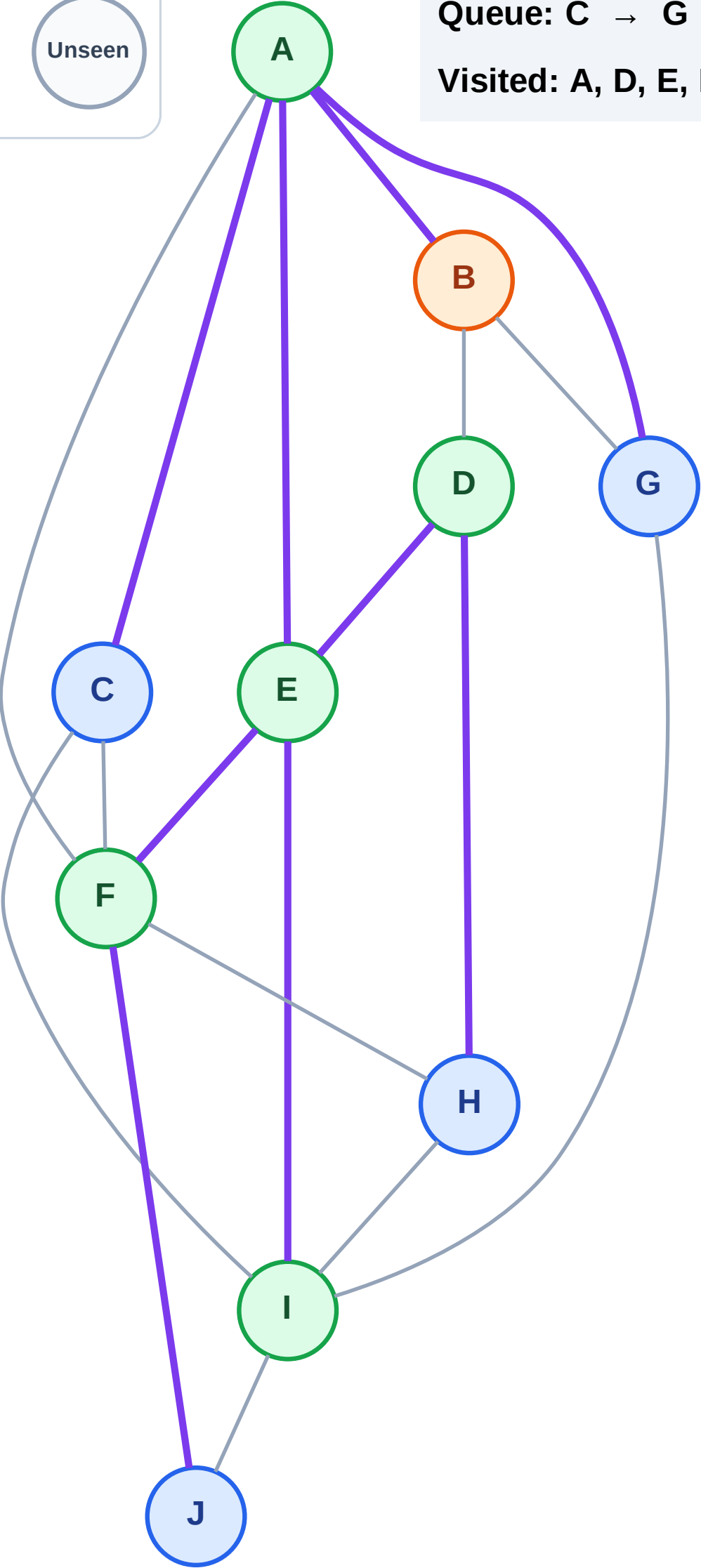
Processing

Discovered

Unseen

Queue: C → G → H → J

Visited: A, D, E, F, I



BFS Traversal

Step 13: No new neighbors from B

Legend

Complete

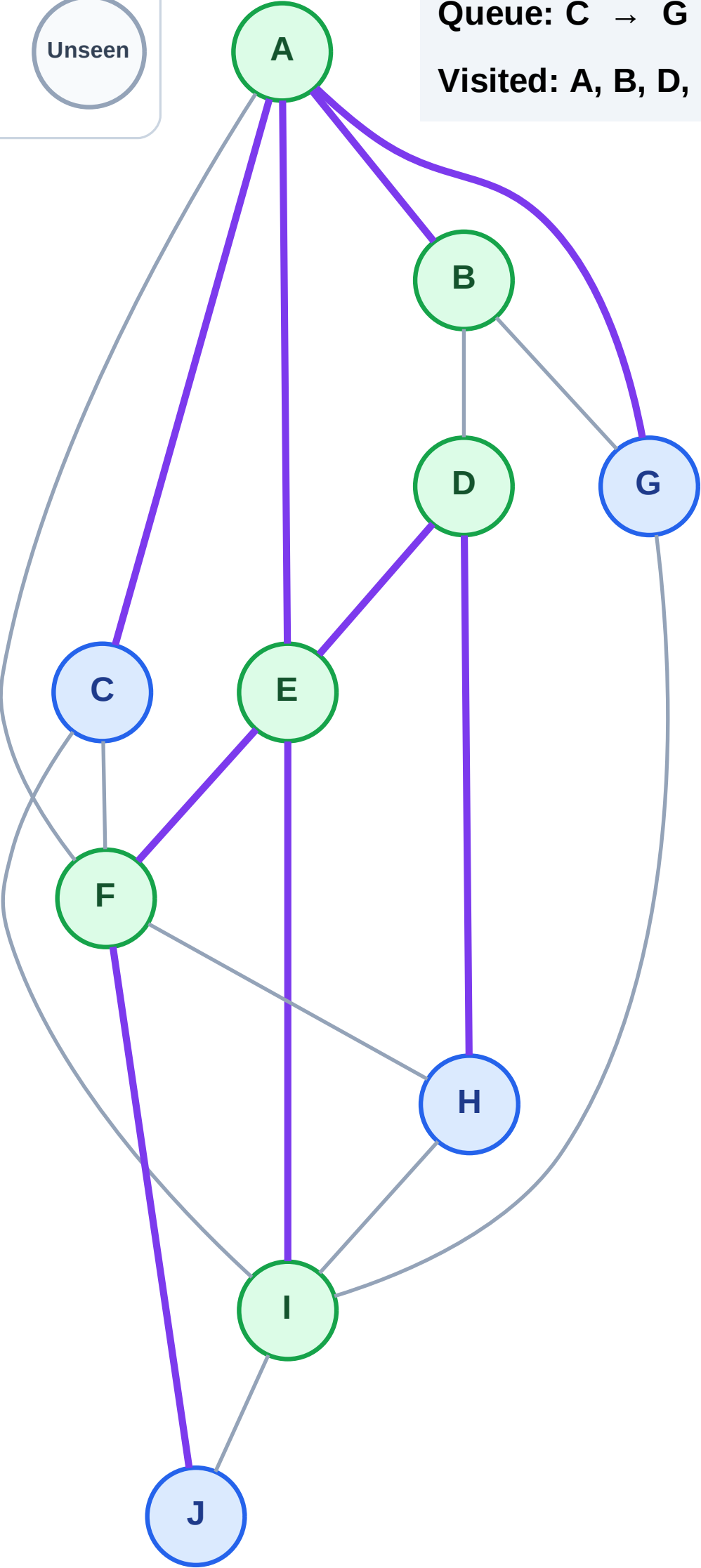
Processing

Discovered

Unseen

Queue: C → G → H → J

Visited: A, B, D, E, F, I



BFS Traversal

Step 14: Process node C

Legend

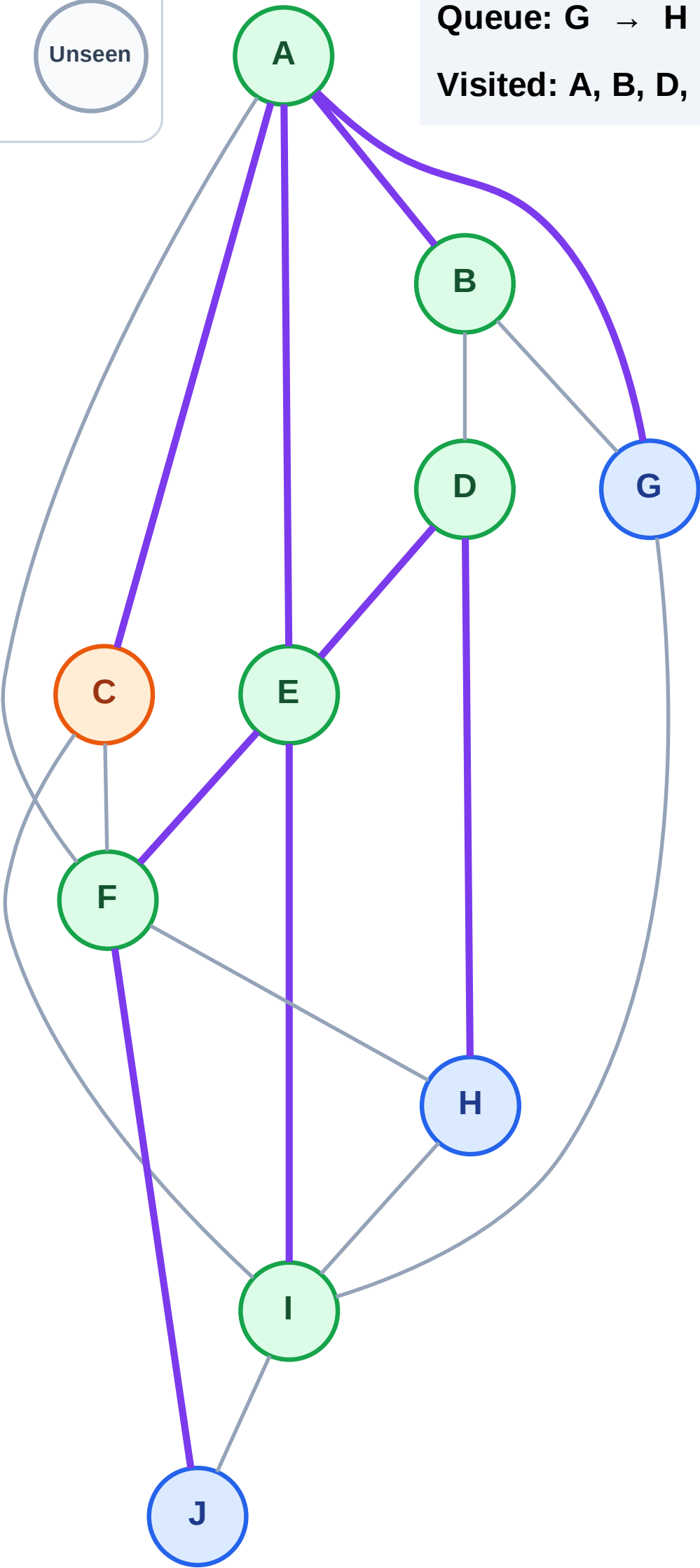
Complete

Processing

Discovered

Unseen

Queue: G → H → J
Visited: A, B, D, E, F, I



BFS Traversal

Step 15: No new neighbors from C

Legend

Complete

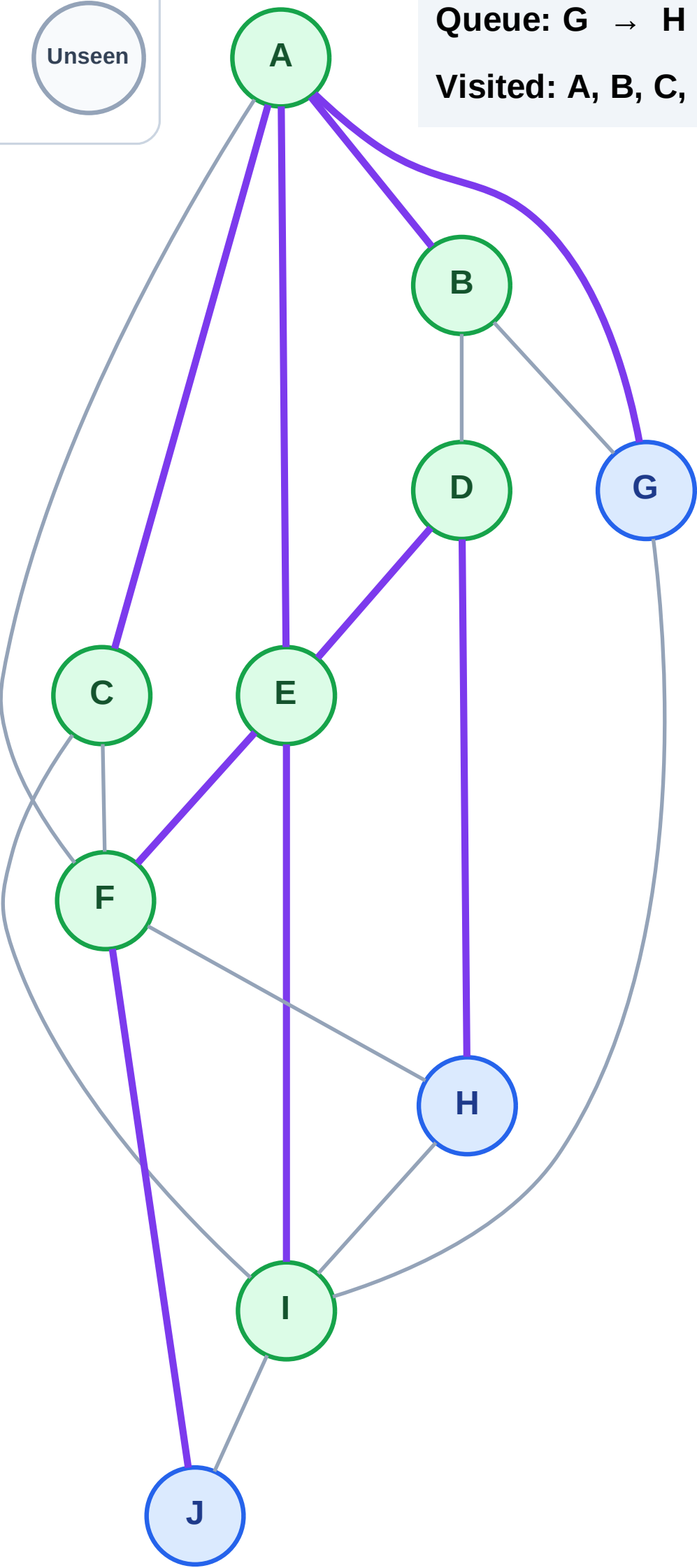
Processing

Discovered

Unseen

Queue: G → H → J

Visited: A, B, C, D, E, F, I



BFS Traversal

Step 16: Process node G

Legend

Complete

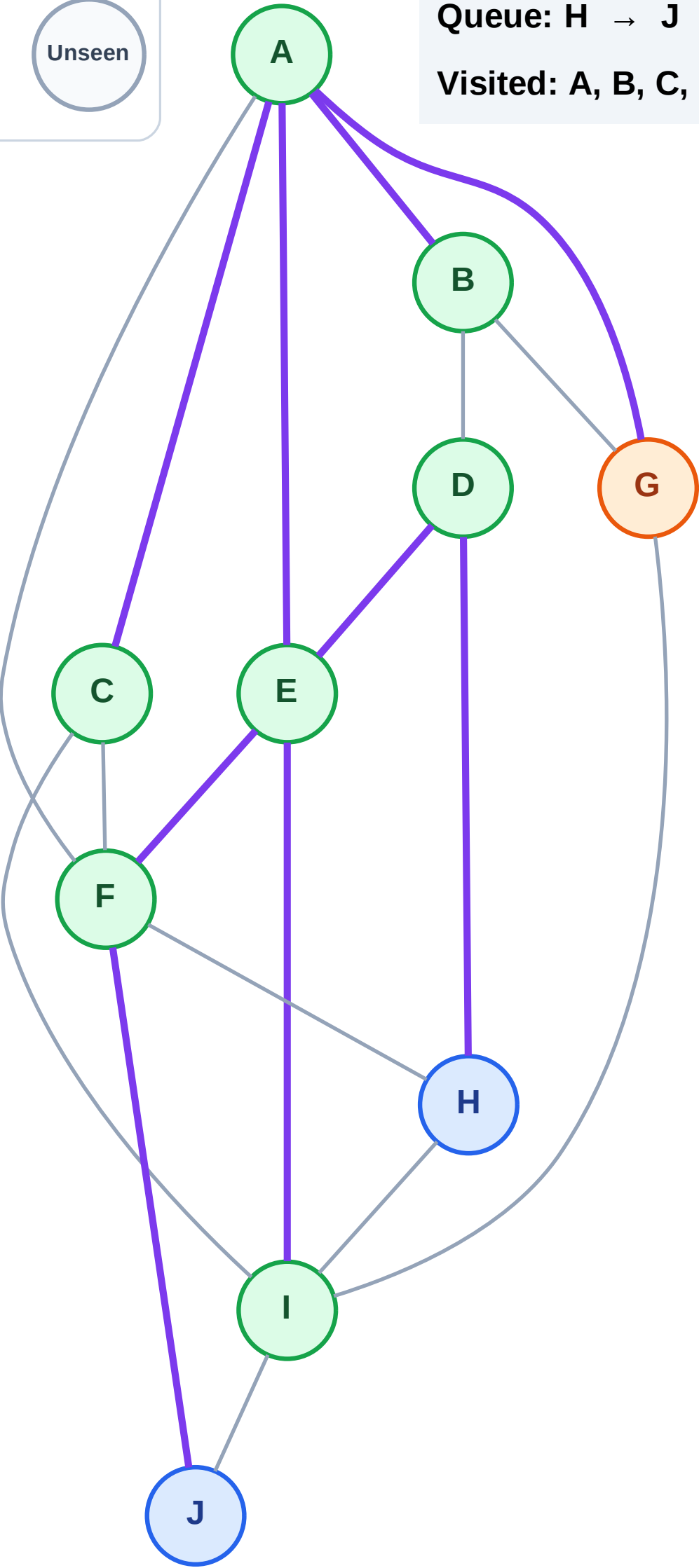
Processing

Discovered

Unseen

Queue: H → J

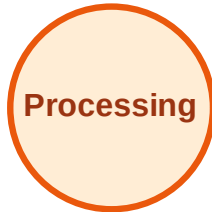
Visited: A, B, C, D, E, F, I



BFS Traversal

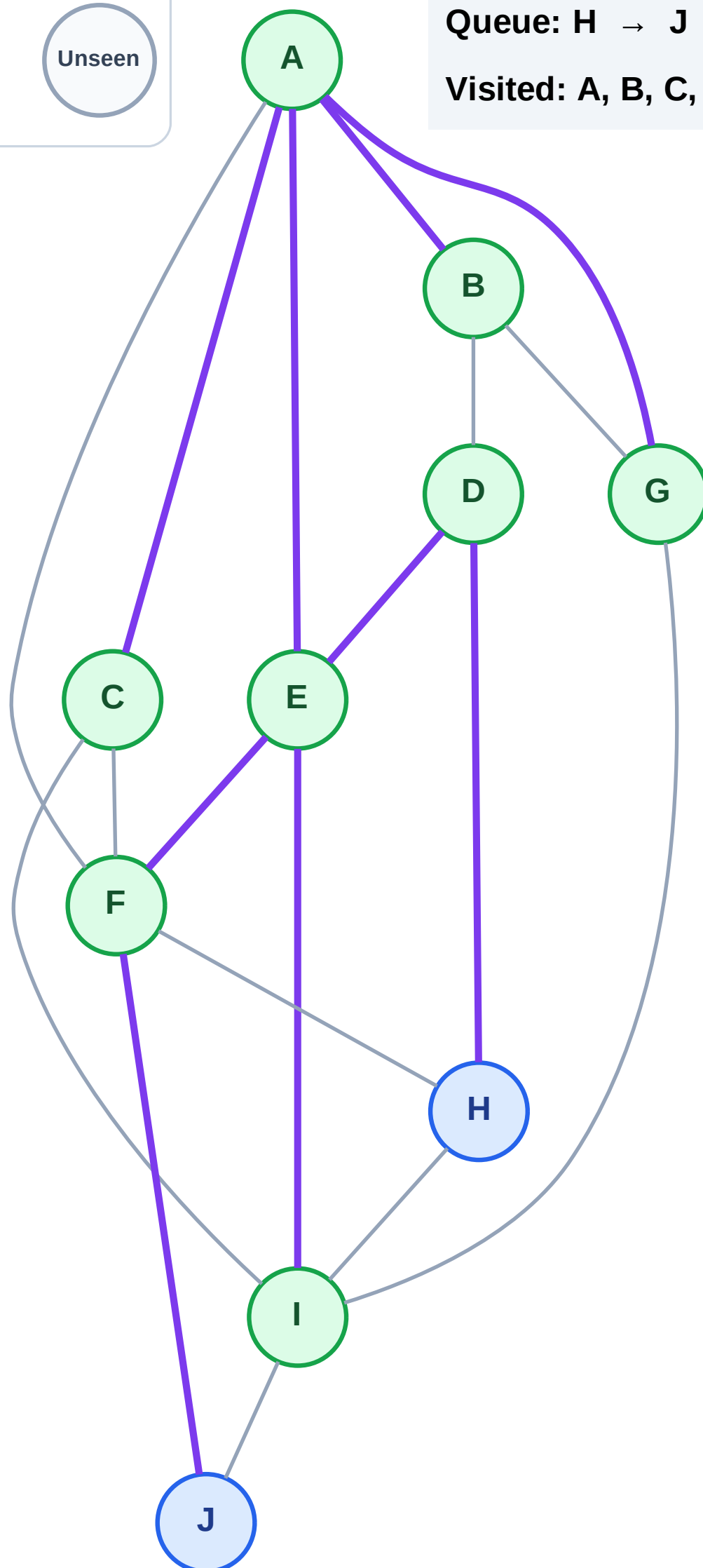
Step 17: No new neighbors from G

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node H

Legend

Complete

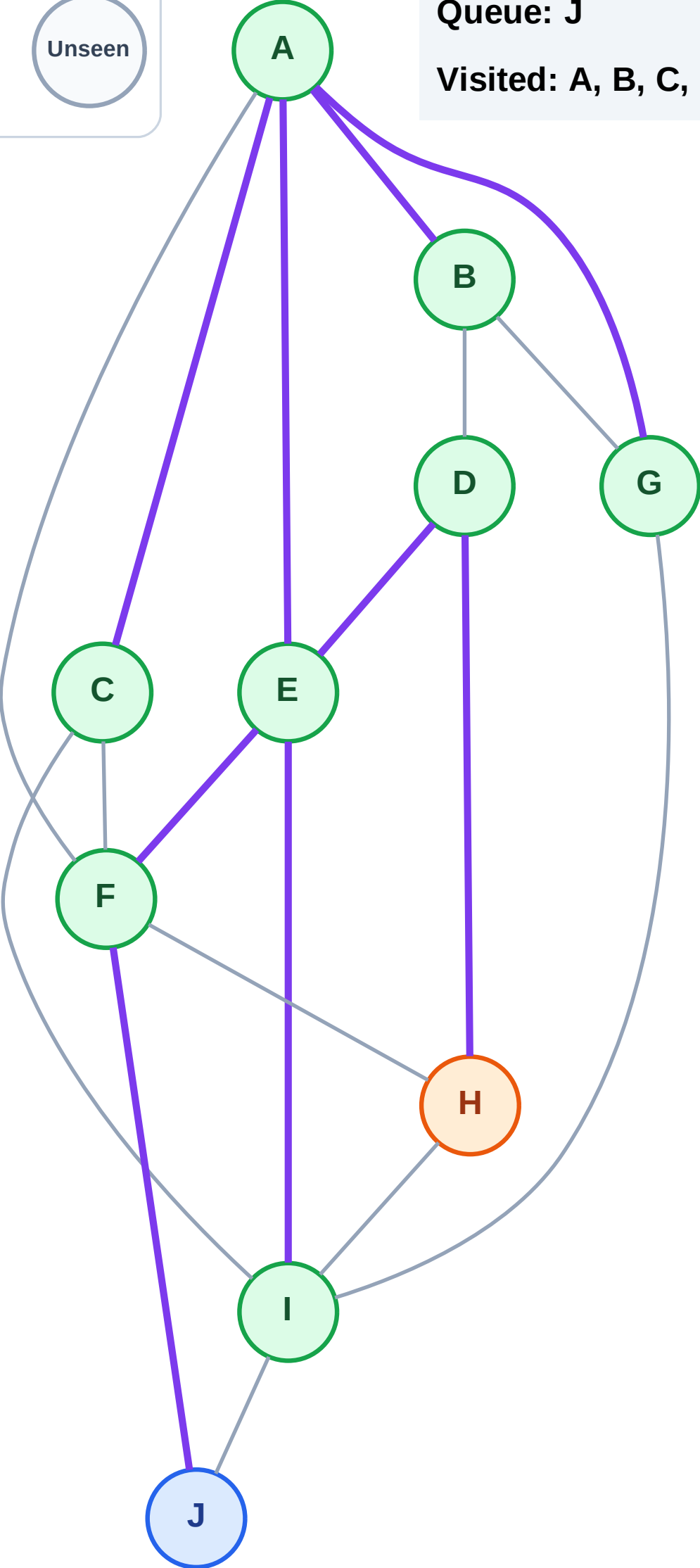
Processing

Discovered

Unseen

Queue: J

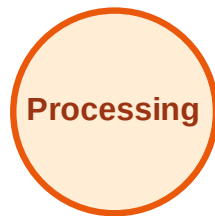
Visited: A, B, C, D, E, F, G, I



BFS Traversal

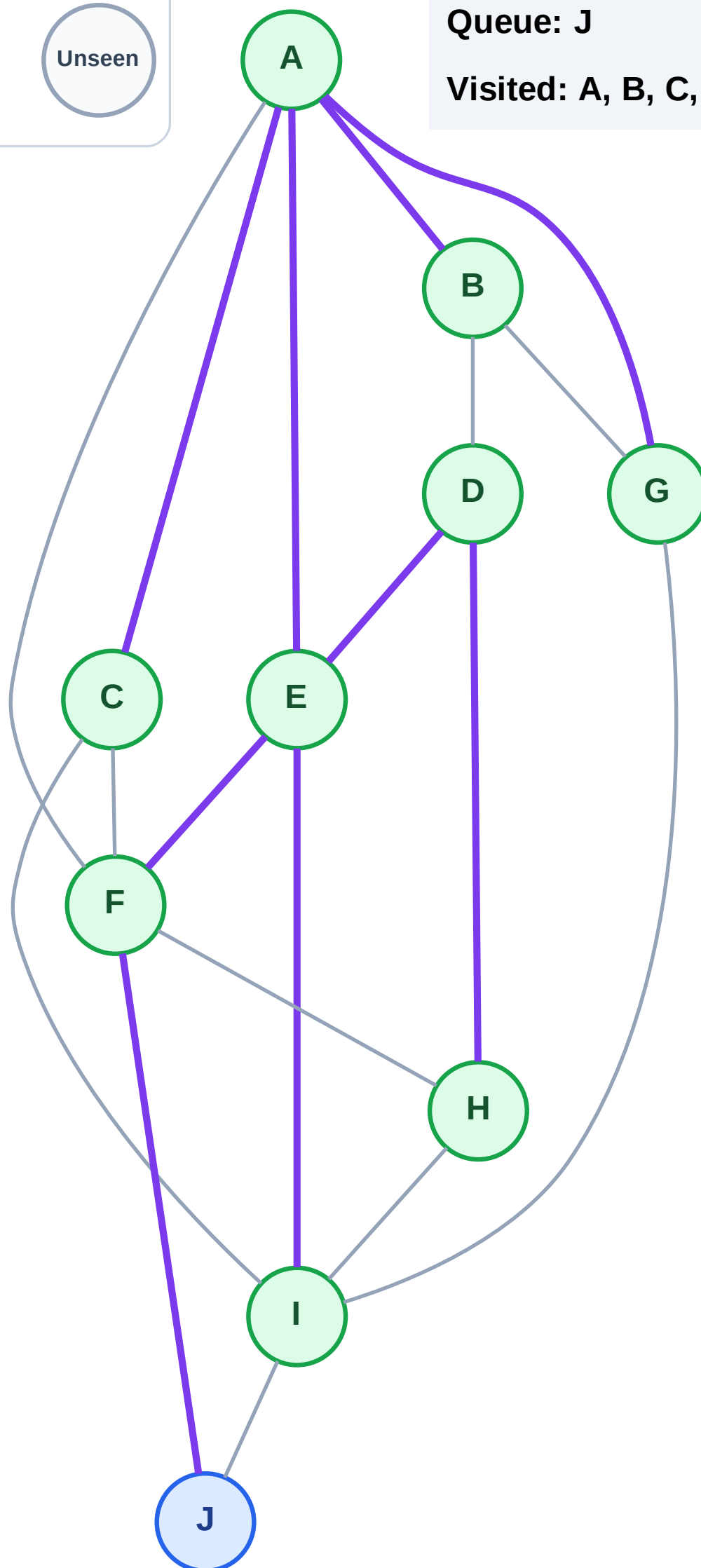
Step 19: No new neighbors from H

Legend



Queue: J

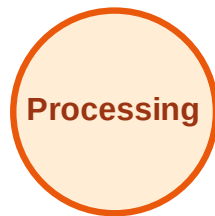
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

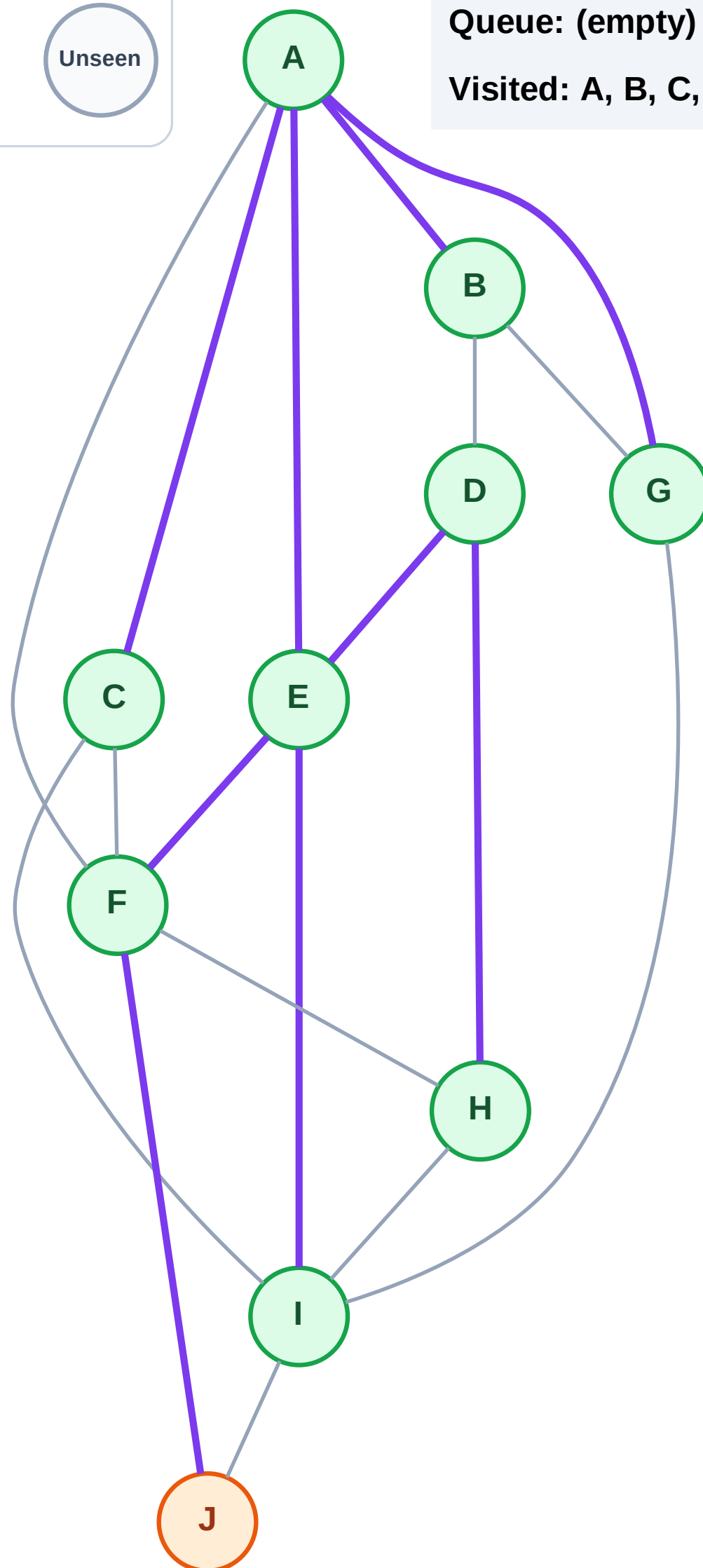
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

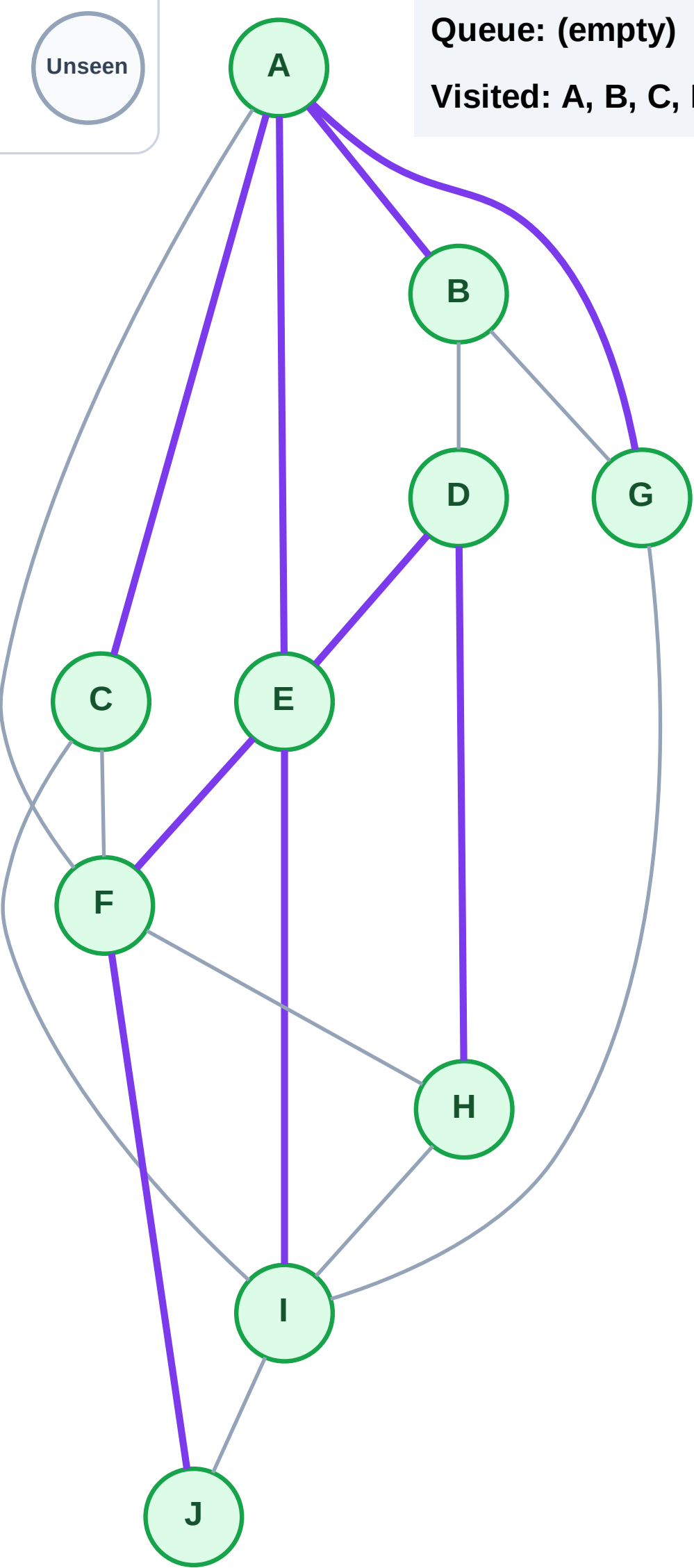
Step 21: No new neighbors from J

Legend



Queue: (empty)

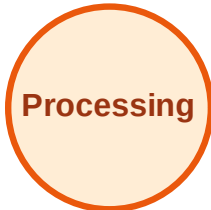
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

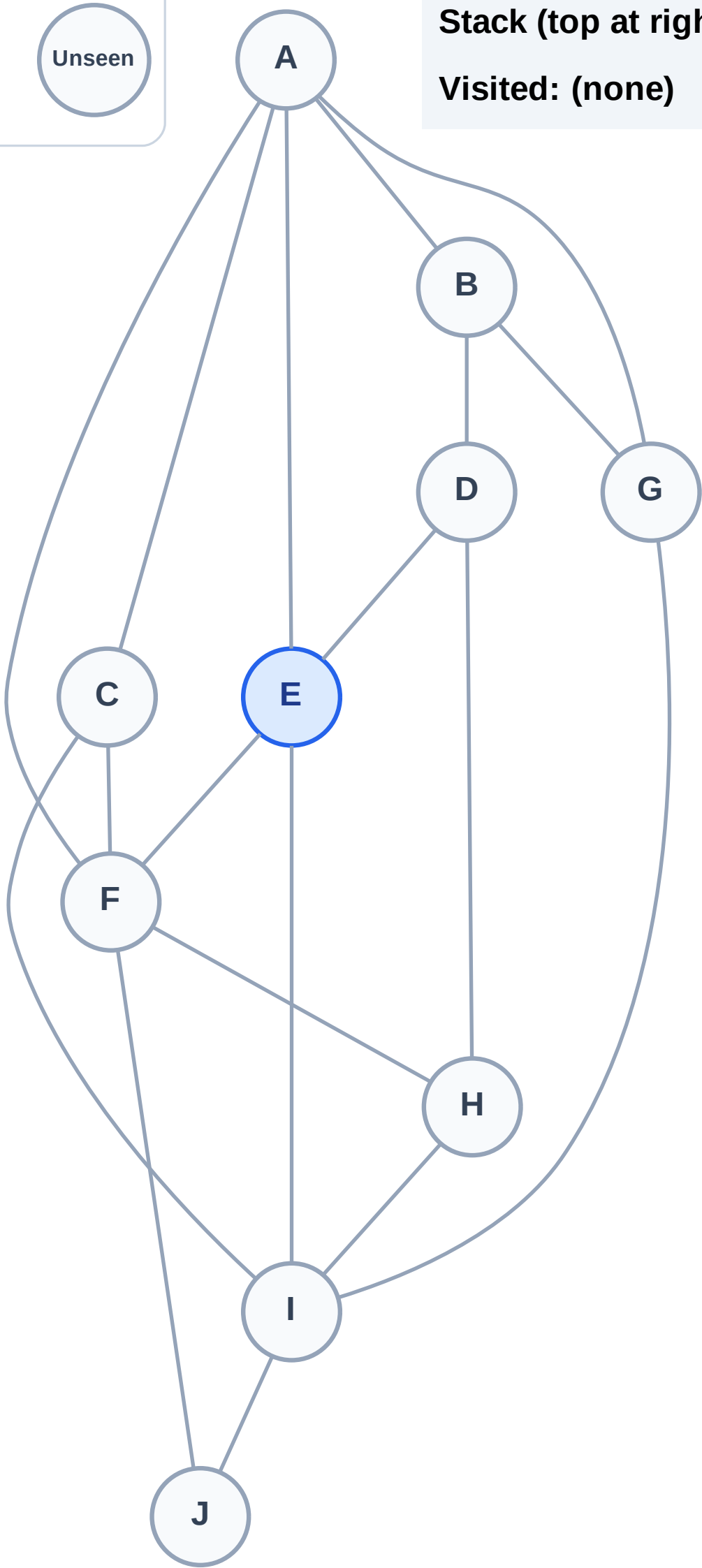
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

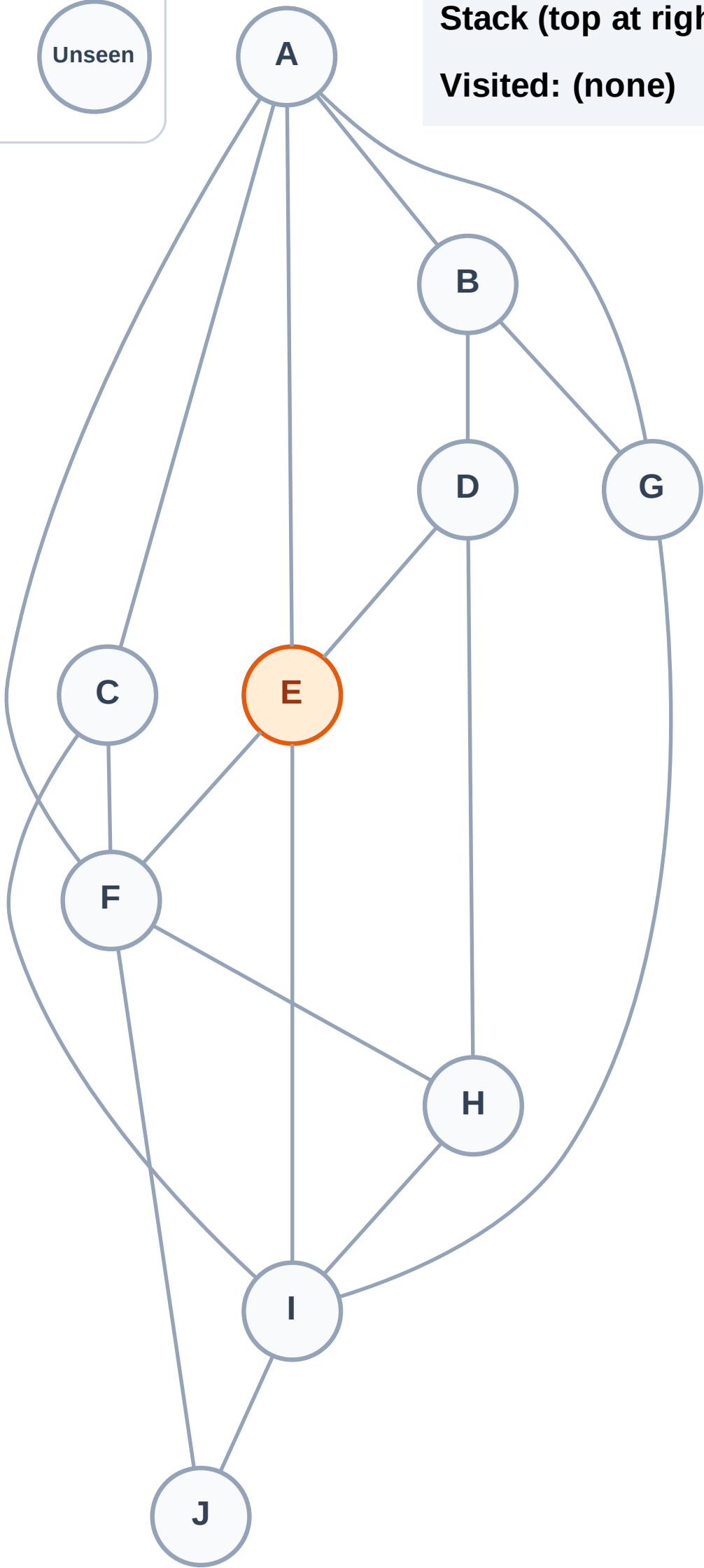
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

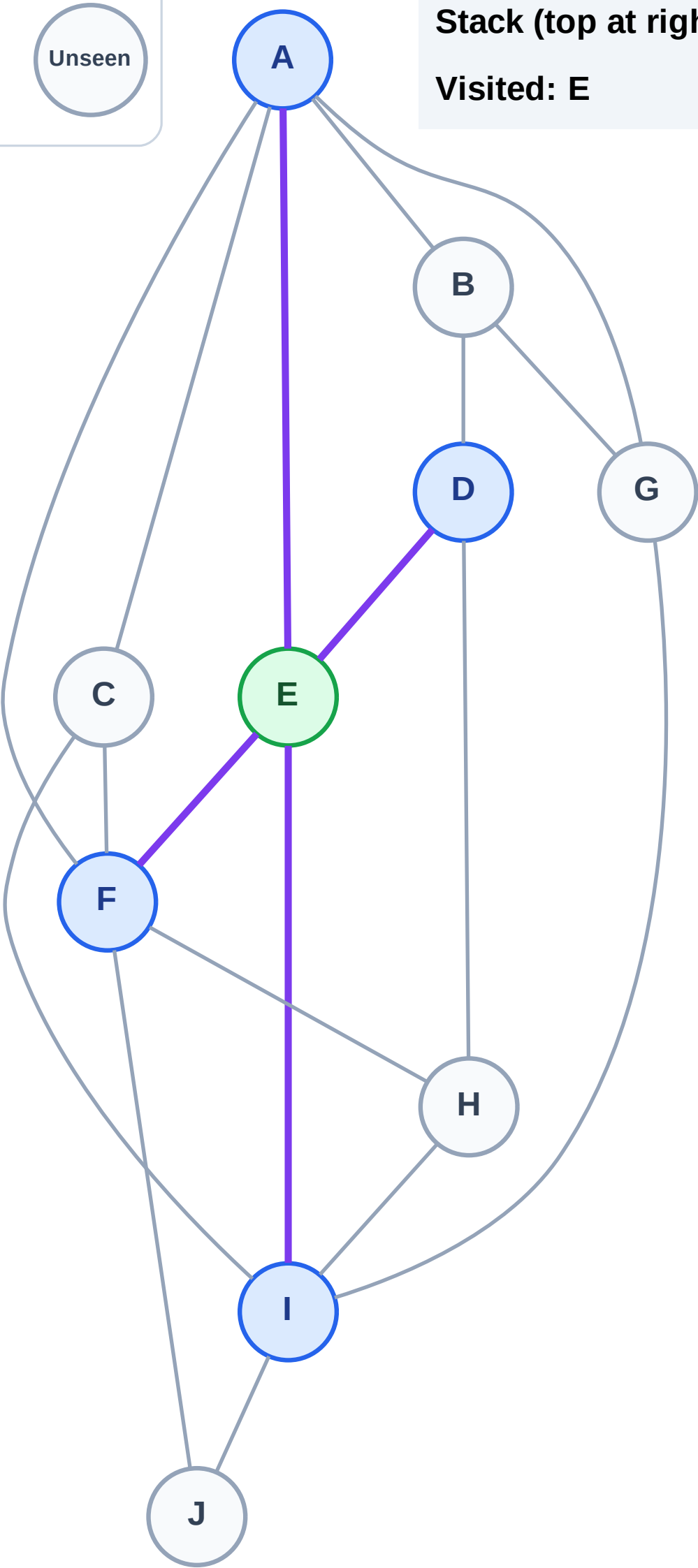
Step 3: Discover I, F, D, A from E

Legend



Stack (top at right): I | F | D | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

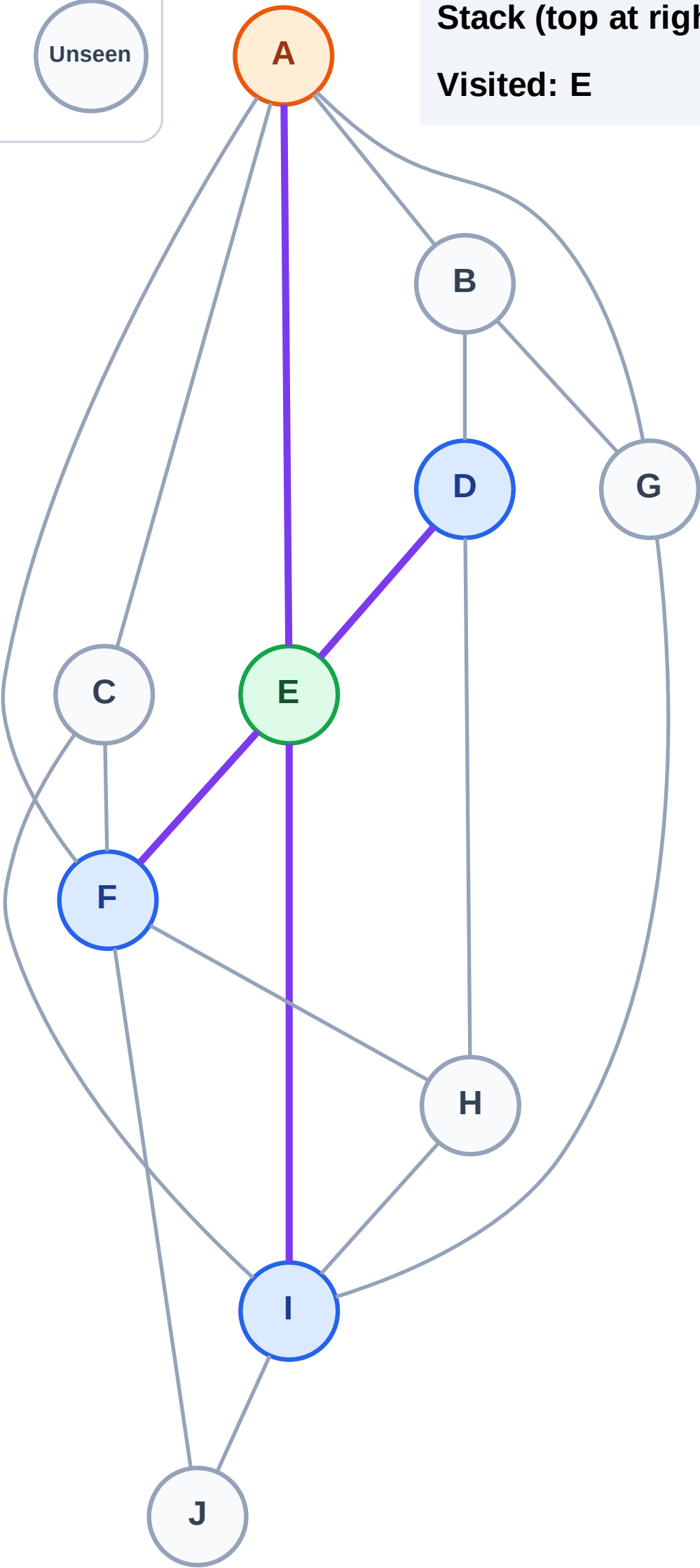
Processing

Discovered

Unseen

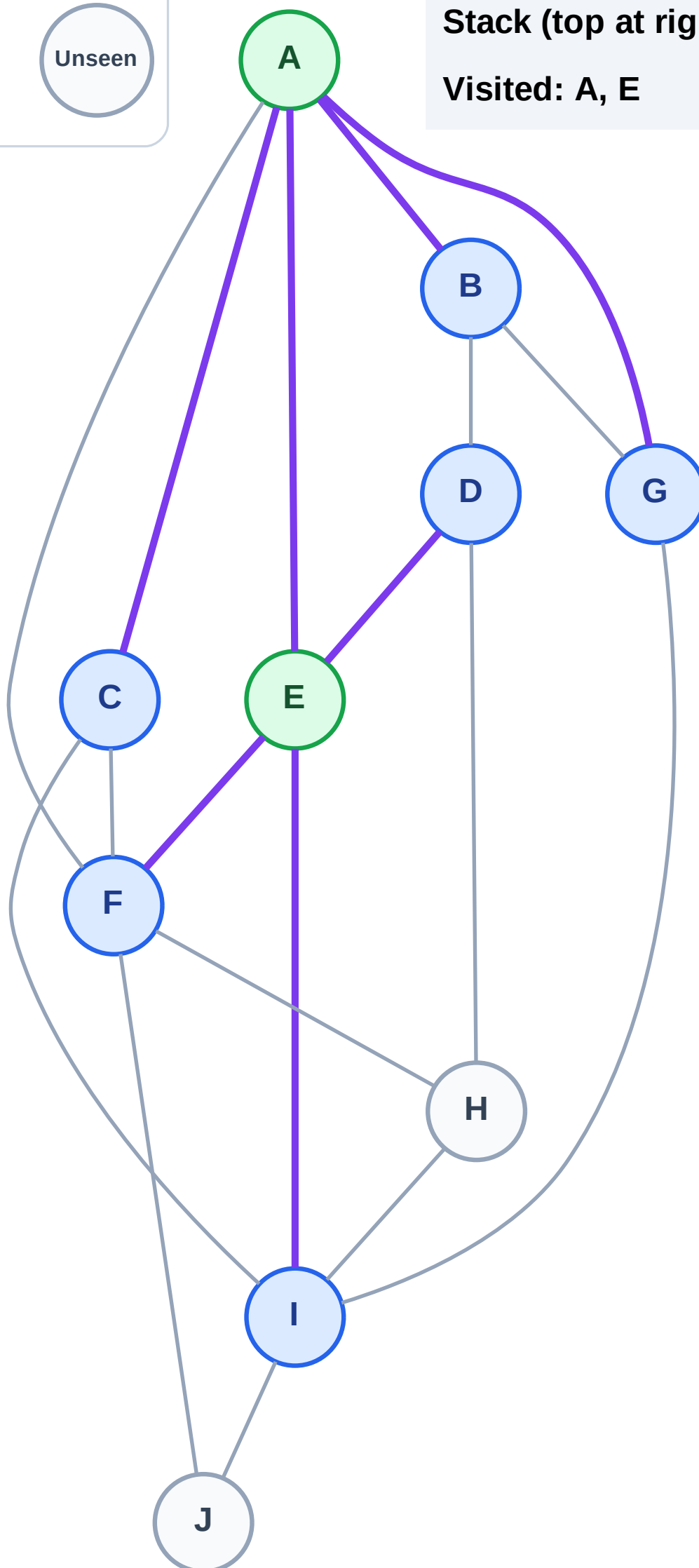
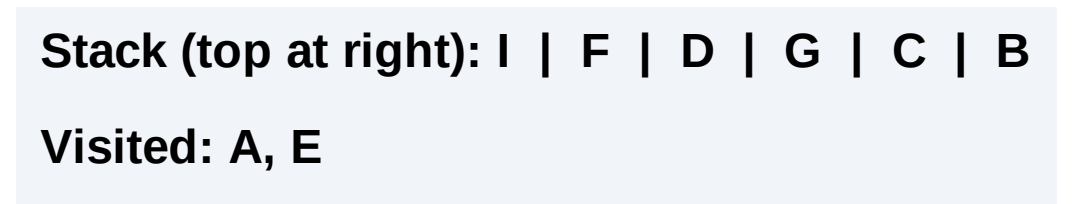
Stack (top at right): I | F | D

Visited: E



Step 5: Discover G, C, B from A

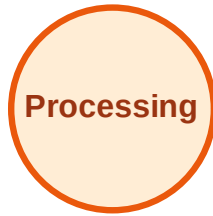
Step 5: Discover G, C, B from A



DFS Traversal

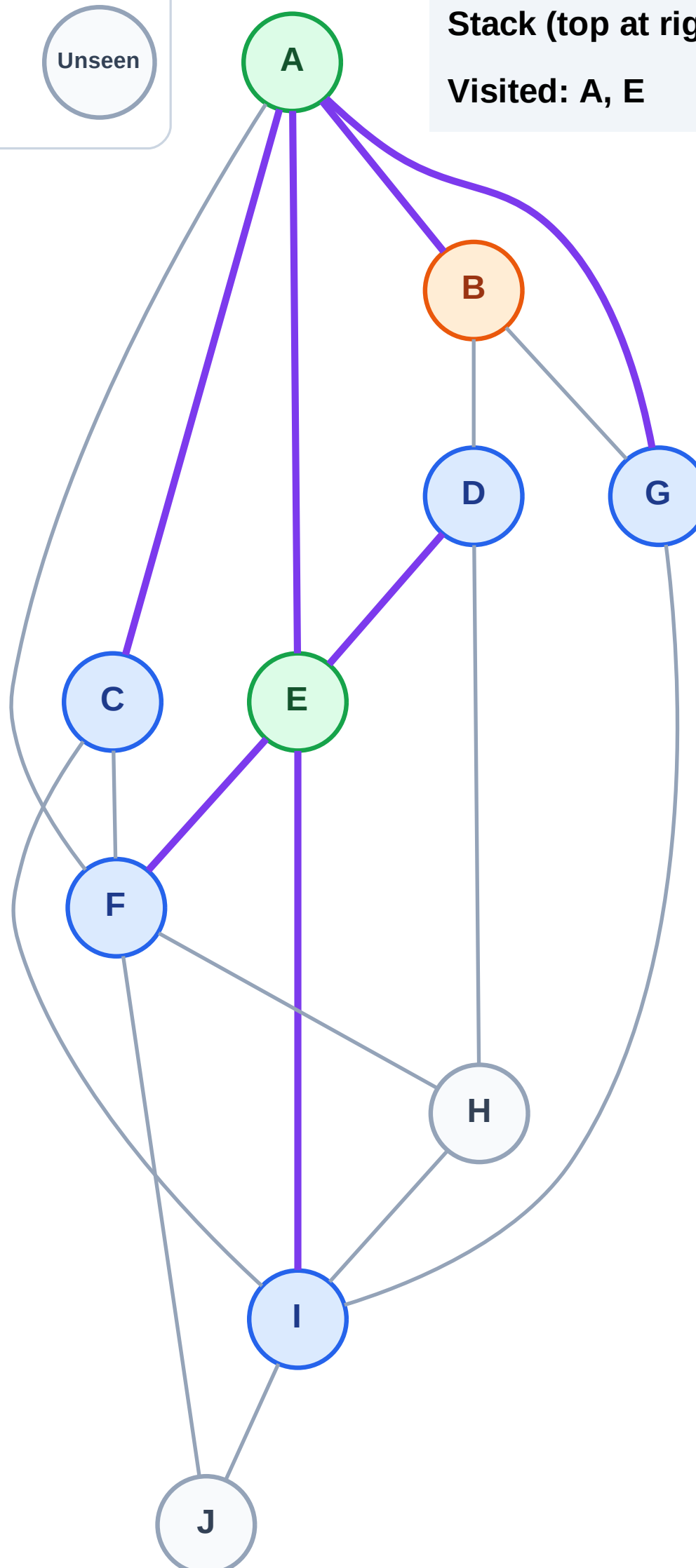
Step 6: Process node B

Legend



Stack (top at right): I | F | D | G | C

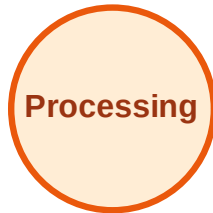
Visited: A, E



DFS Traversal

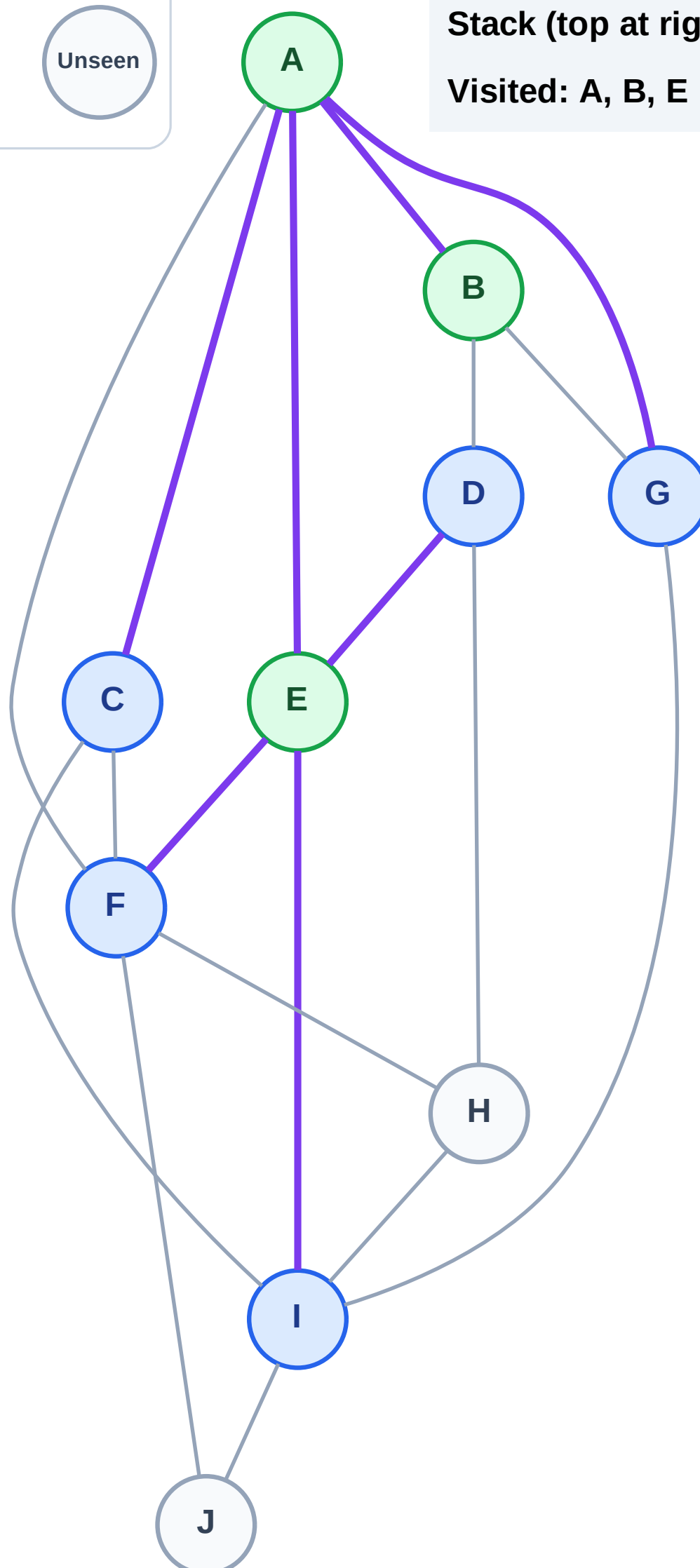
Step 7: No new neighbors from B

Legend



Stack (top at right): I | F | D | G | C

Visited: A, B, E



DFS Traversal

Step 8: Process node C

Legend

Complete

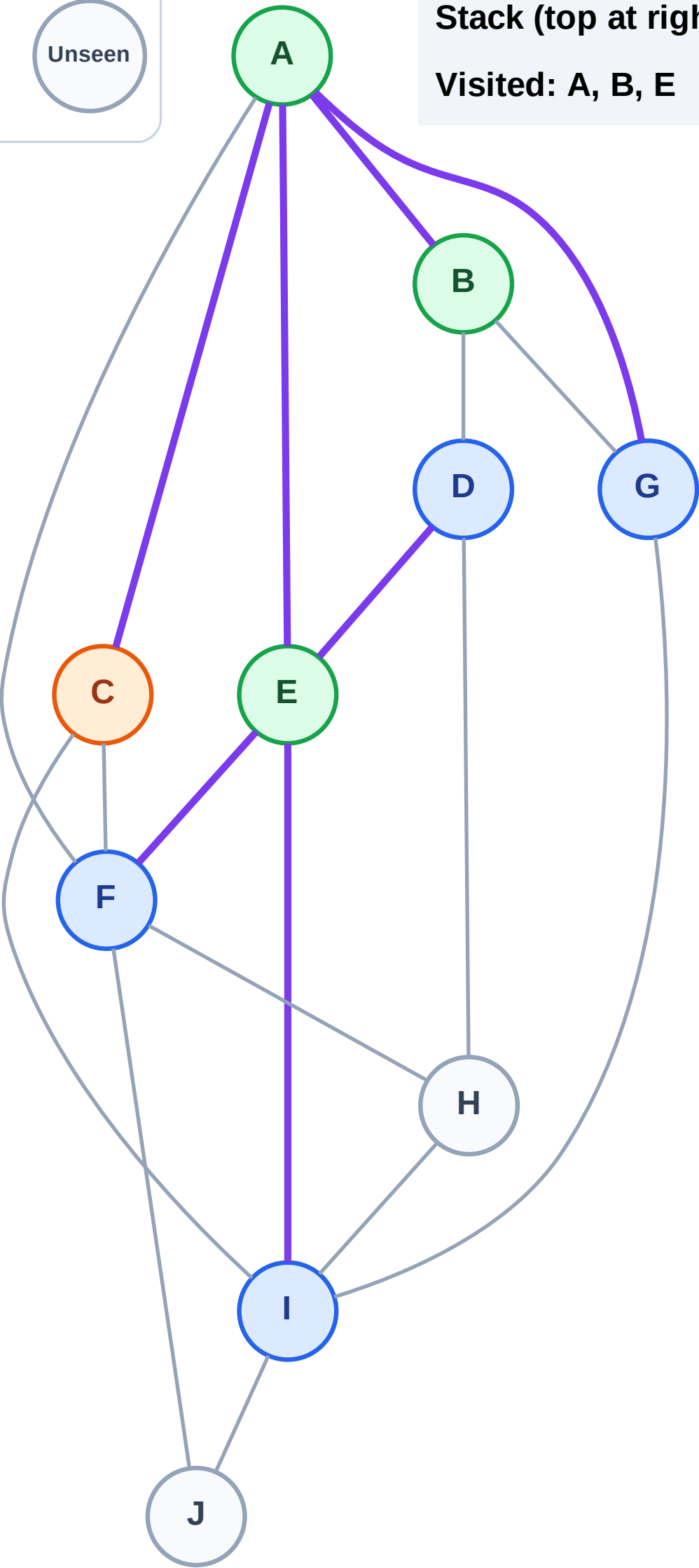
Processing

Discovered

Unseen

Stack (top at right): I | F | D | G

Visited: A, B, E



DFS Traversal

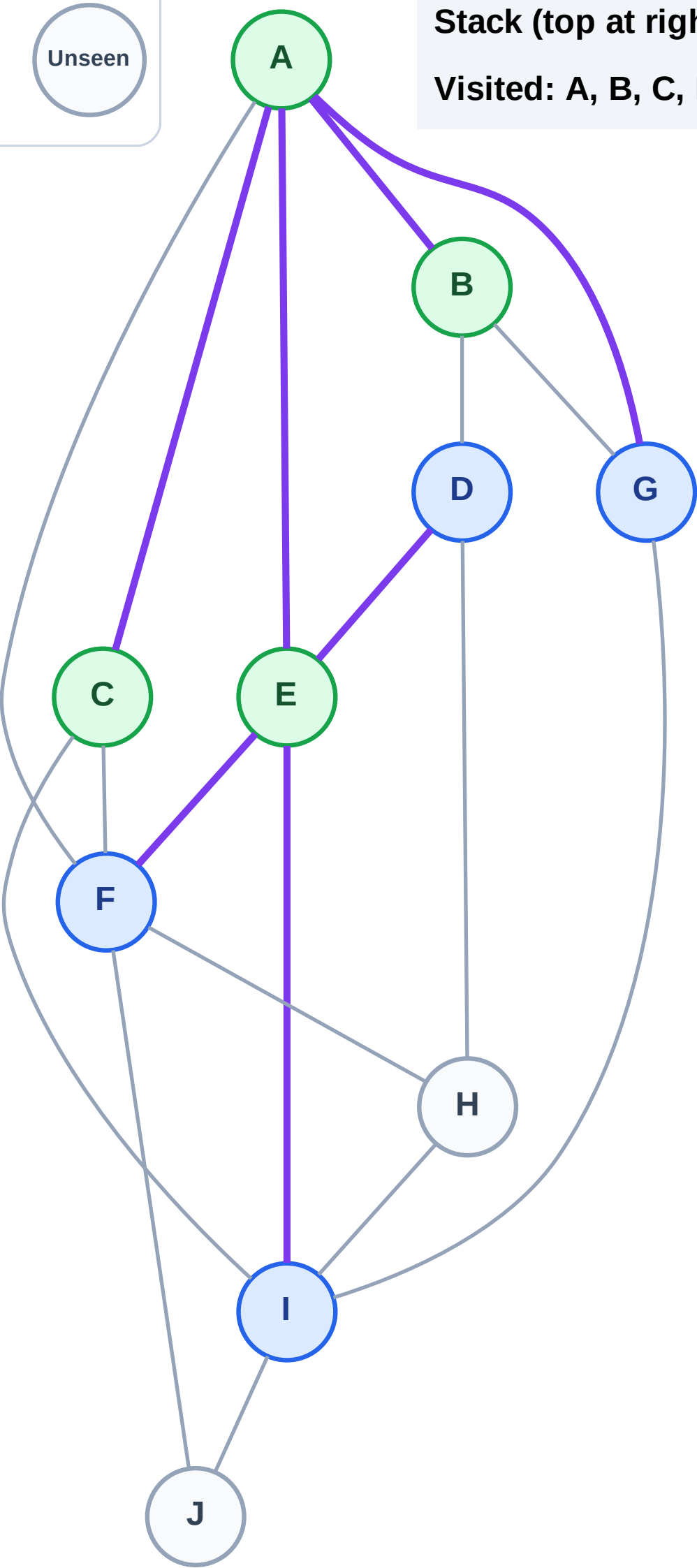
Step 9: No new neighbors from C

Legend



Stack (top at right): I | F | D | G

Visited: A, B, C, E



DFS Traversal

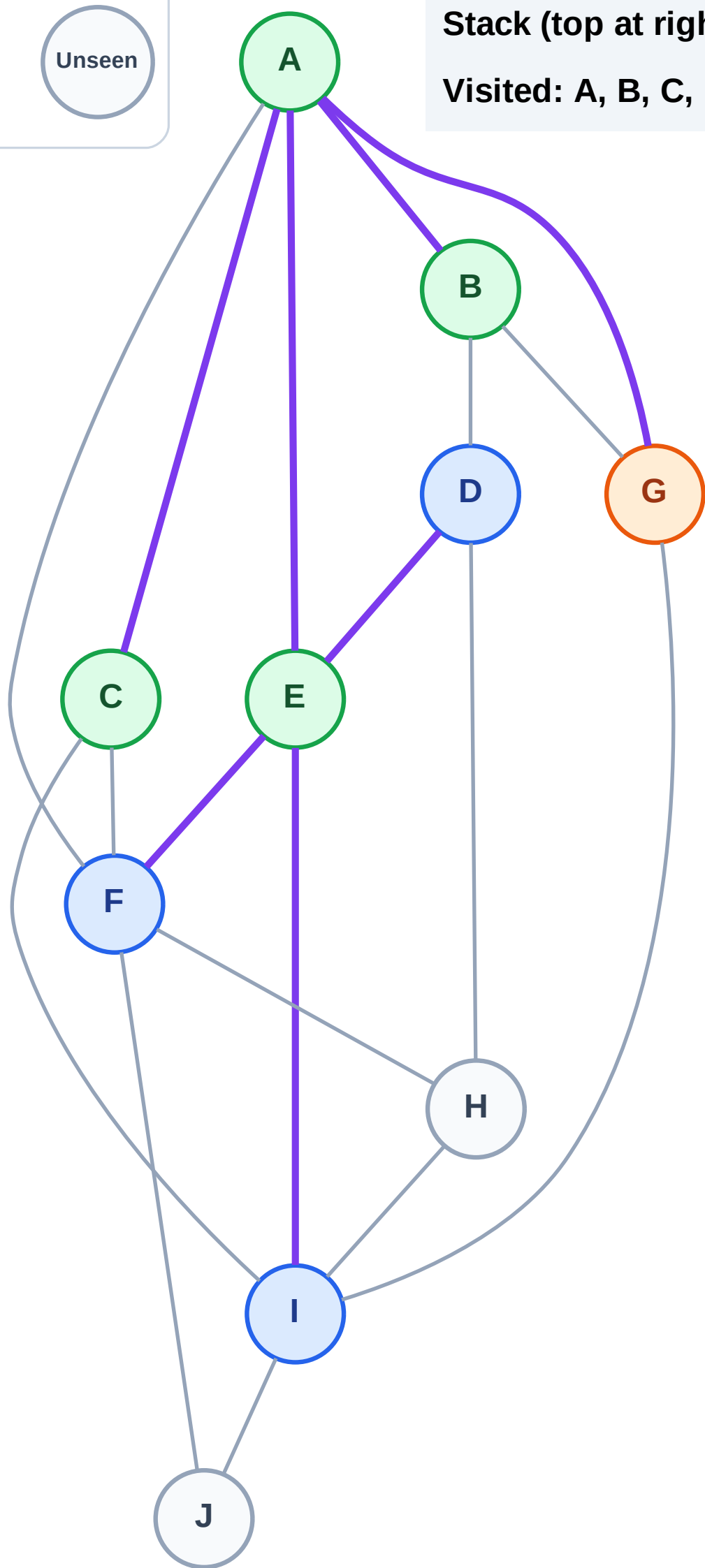
Step 10: Process node G

Legend



Stack (top at right): I | F | D

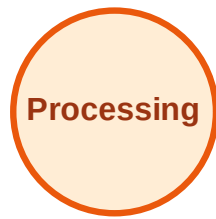
Visited: A, B, C, E



DFS Traversal

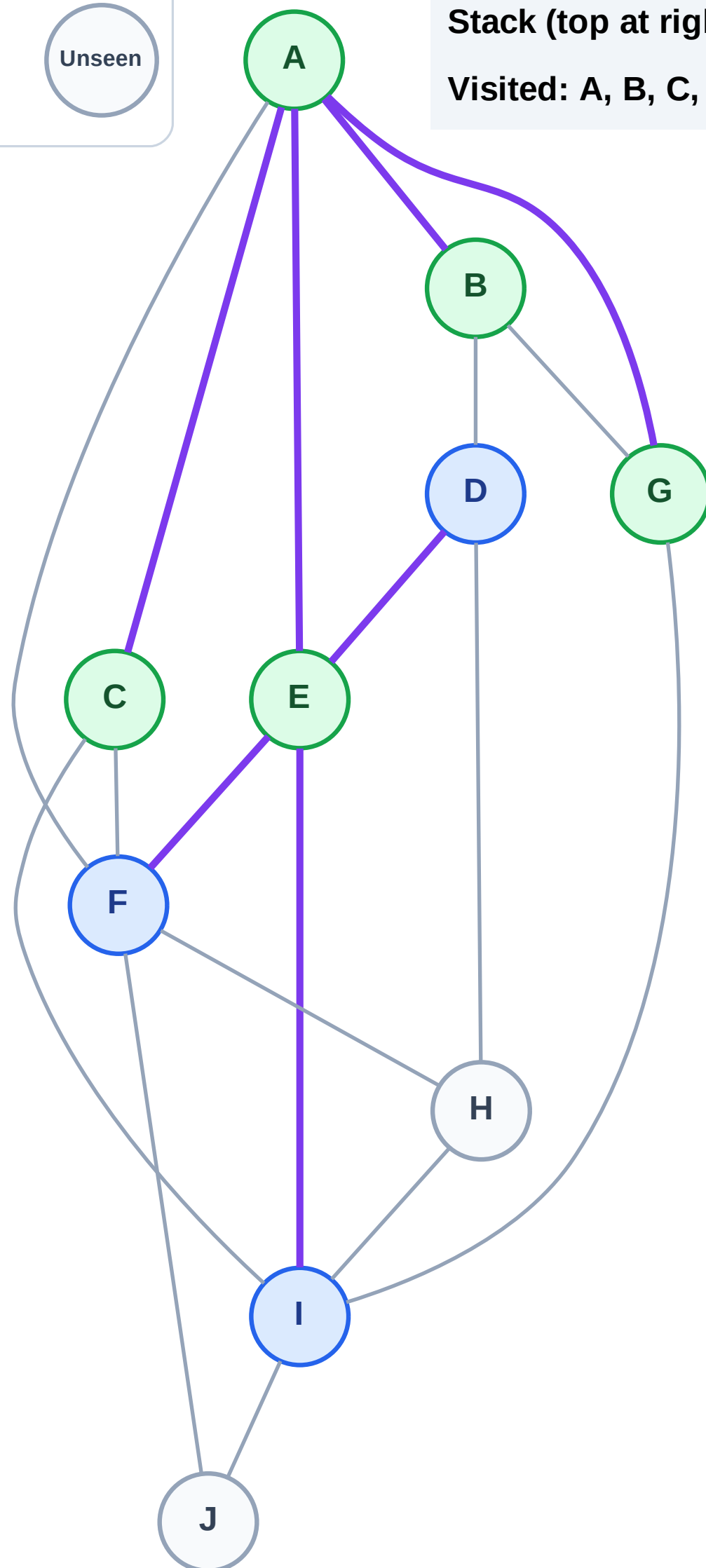
Step 11: No new neighbors from G

Legend



Stack (top at right): I | F | D

Visited: A, B, C, E, G



DFS Traversal

Step 12: Process node D

Legend

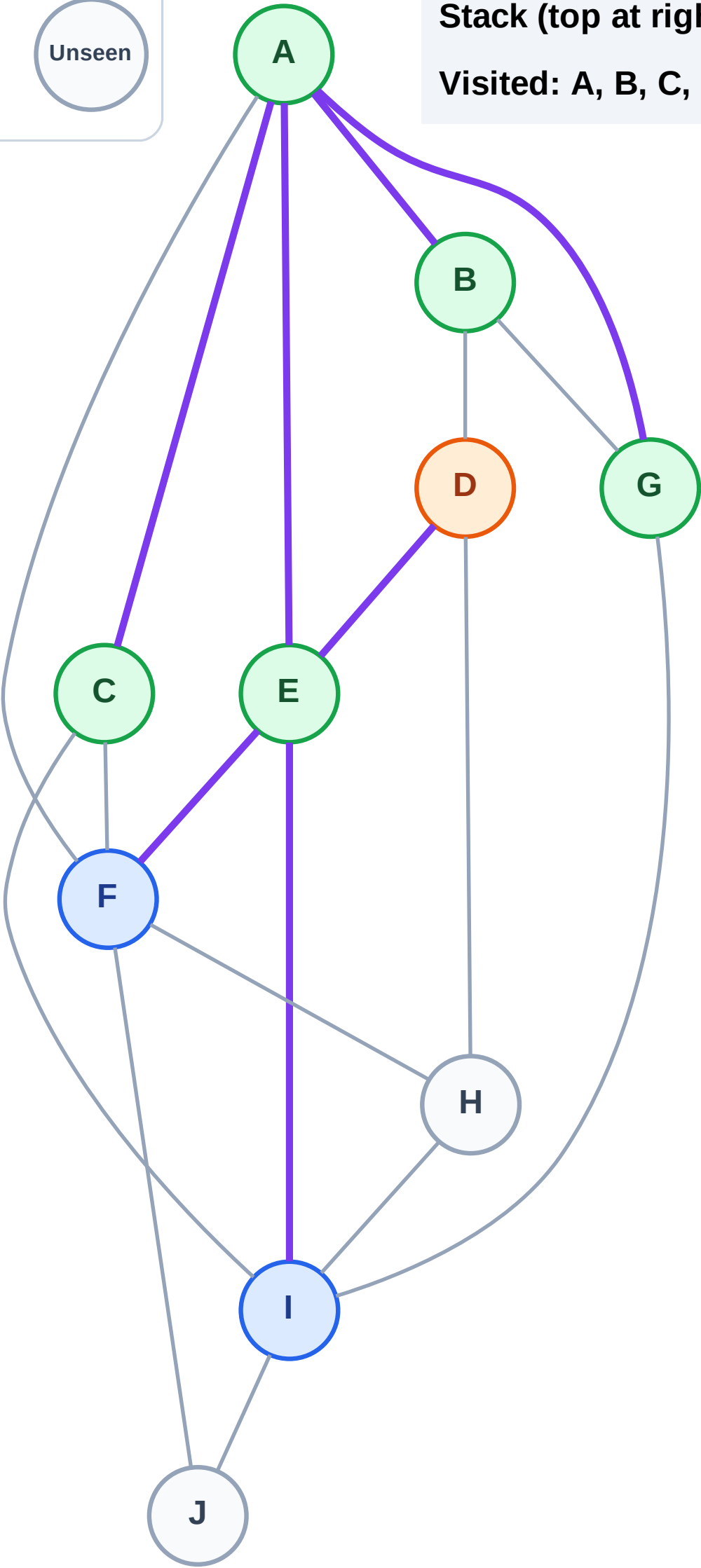
Complete

Processing

Discovered

Unseen

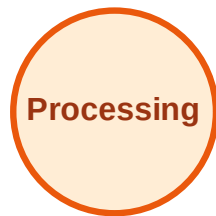
Stack (top at right): I | F
Visited: A, B, C, E, G



DFS Traversal

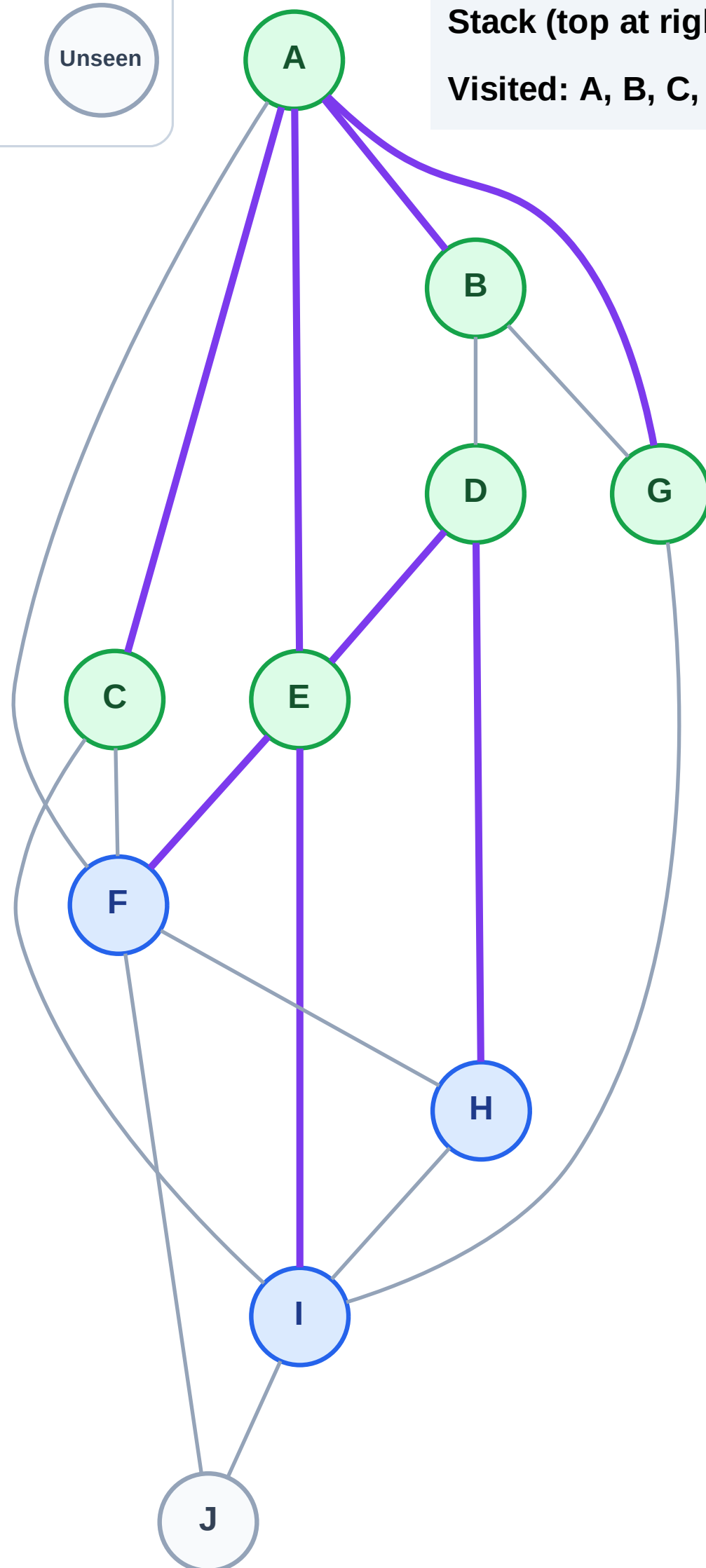
Step 13: Discover H from D

Legend



Stack (top at right): I | F | H

Visited: A, B, C, D, E, G



DFS Traversal

Step 14: Process node H

Legend

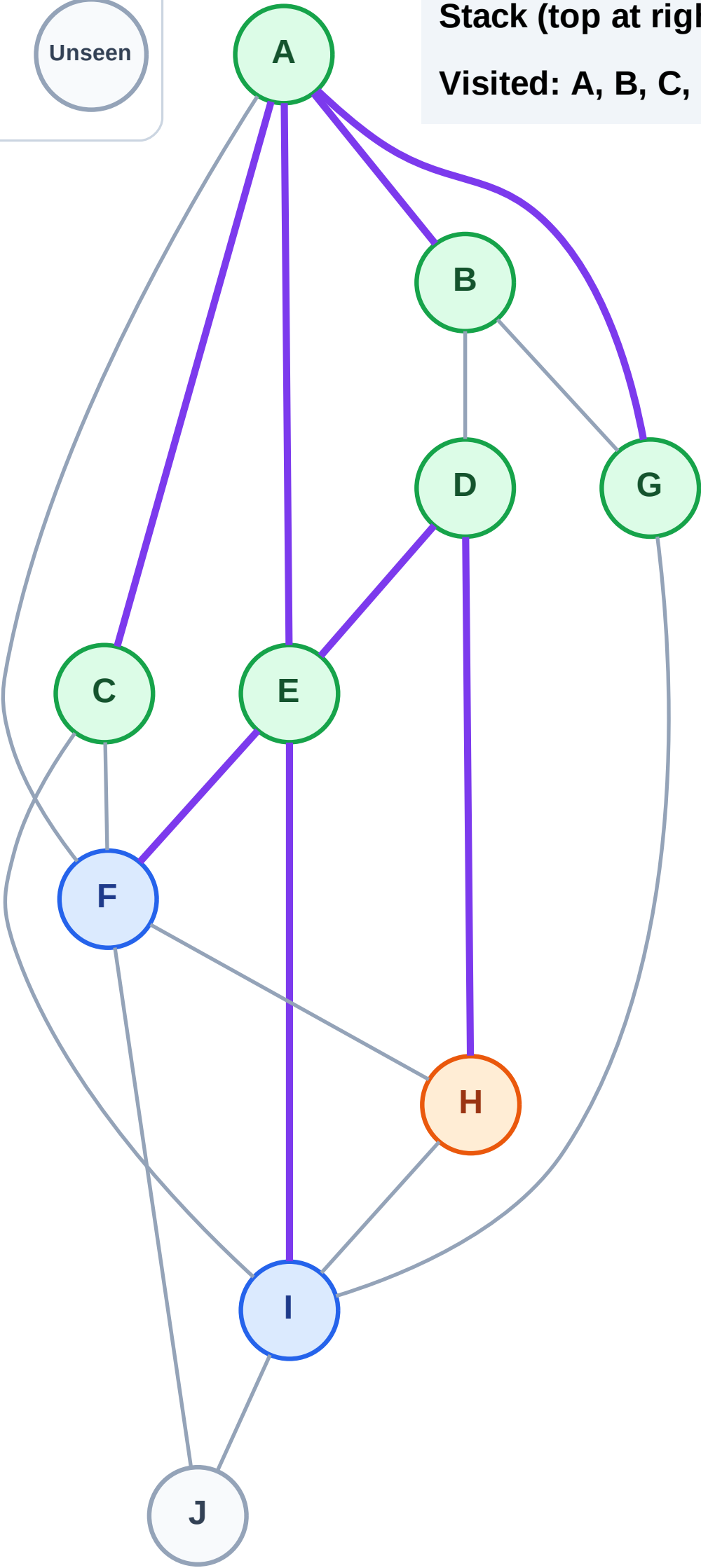
Complete

Processing

Discovered

Unseen

Stack (top at right): I | F
Visited: A, B, C, D, E, G



DFS Traversal

Step 15: No new neighbors from H

Legend

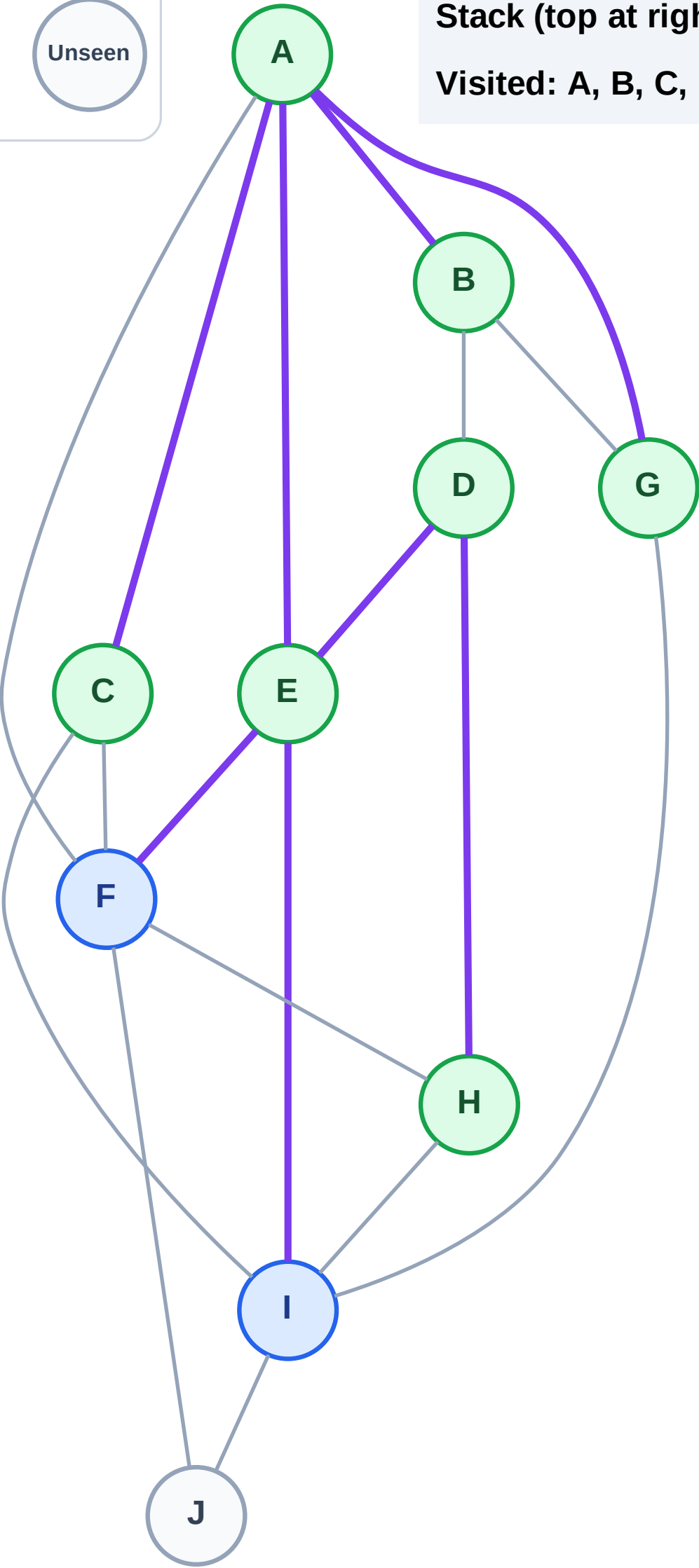
Complete

Processing

Discovered

Unseen

Stack (top at right): I | F
Visited: A, B, C, D, E, G, H



DFS Traversal

Step 16: Process node F

Legend

Complete

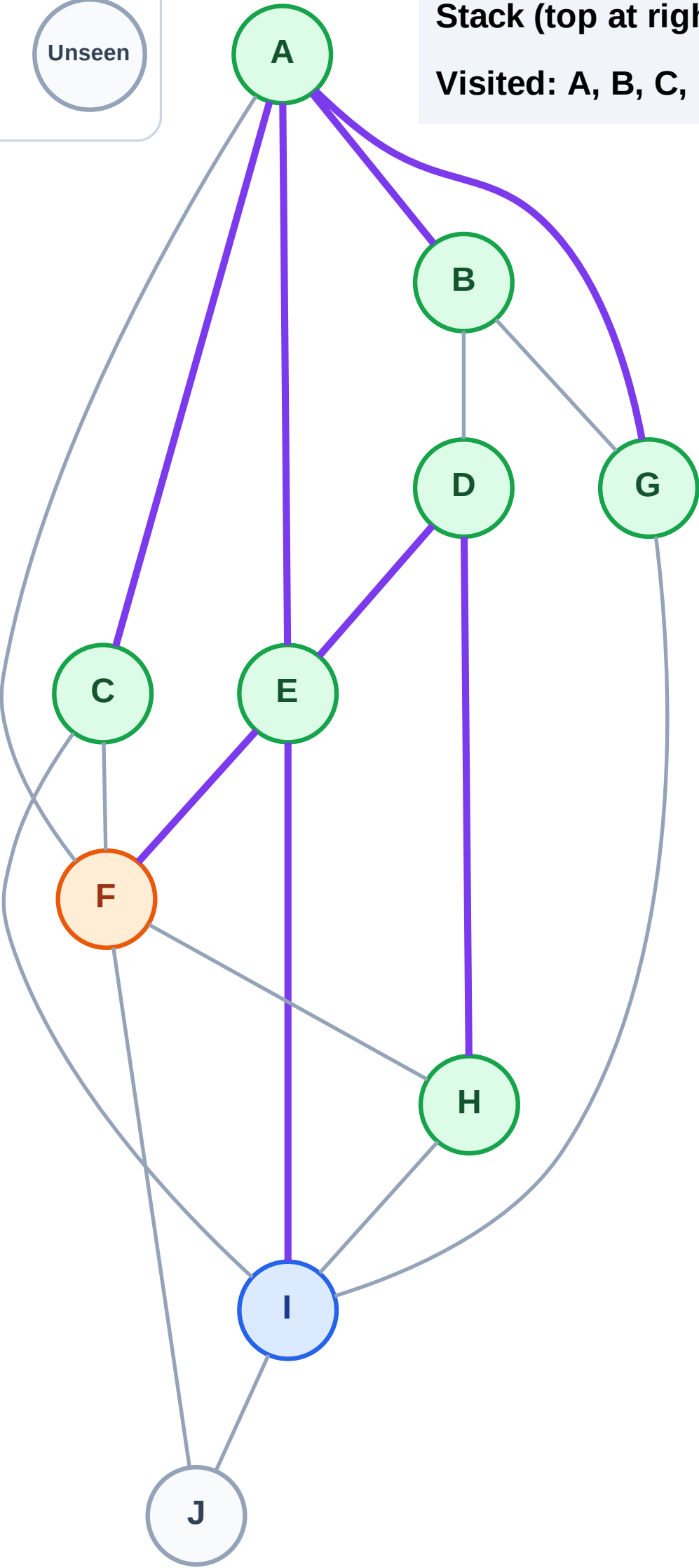
Processing

Discovered

Unseen

Stack (top at right): I

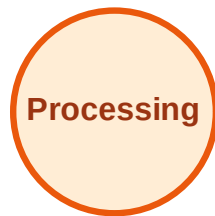
Visited: A, B, C, D, E, G, H



DFS Traversal

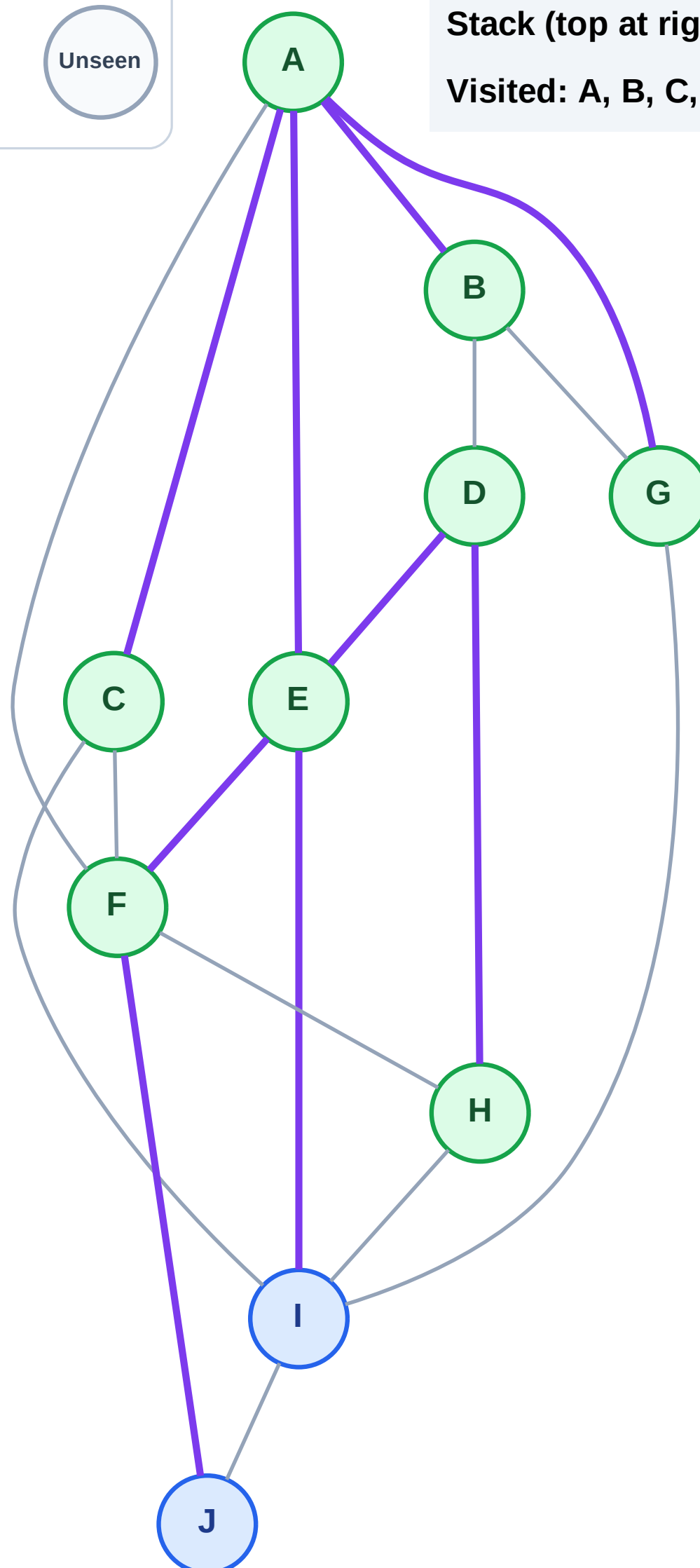
Step 17: Discover J from F

Legend



Stack (top at right): I | J

Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 18: Process node J

Legend

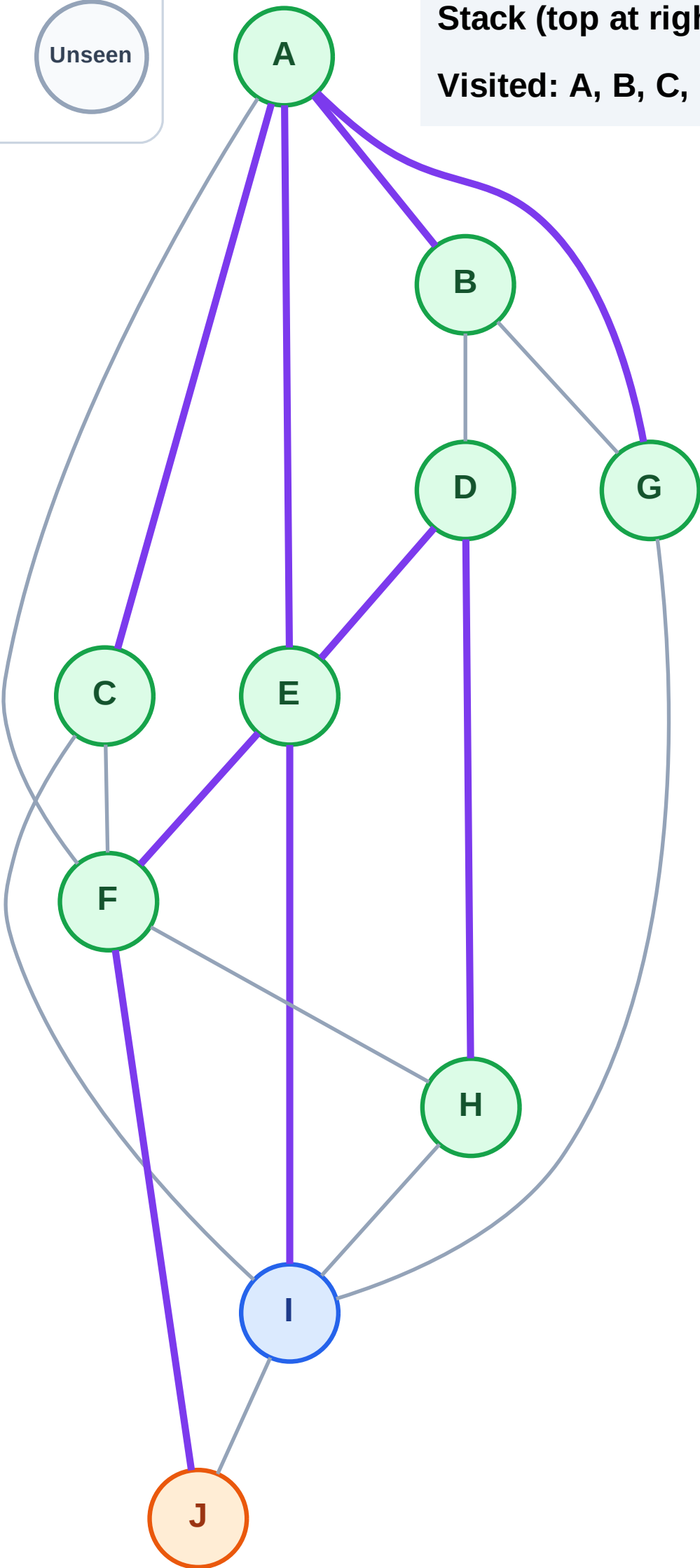
Complete

Processing

Discovered

Unseen

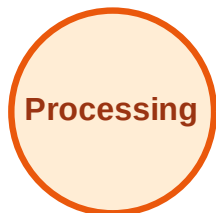
Stack (top at right): I
Visited: A, B, C, D, E, F, G, H



DFS Traversal

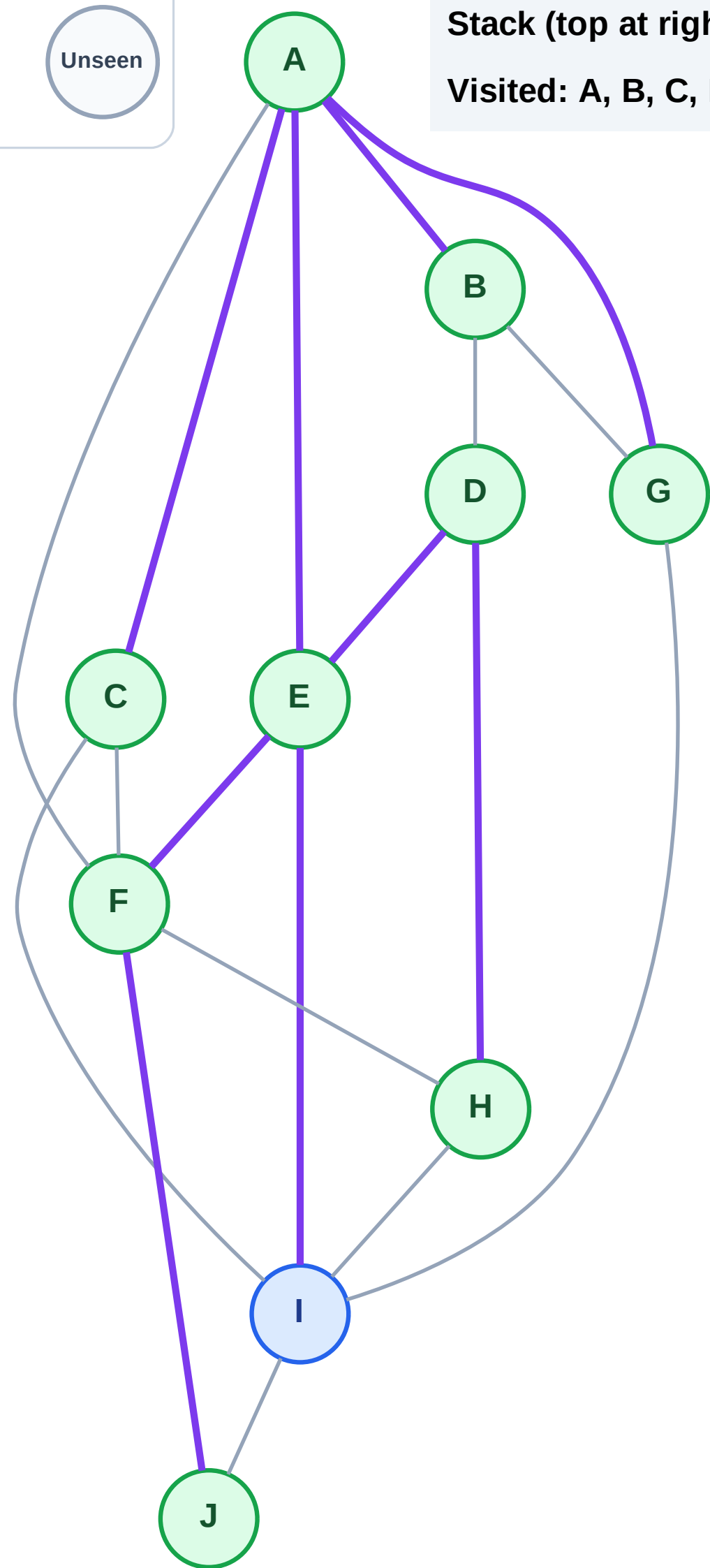
Step 19: No new neighbors from J

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

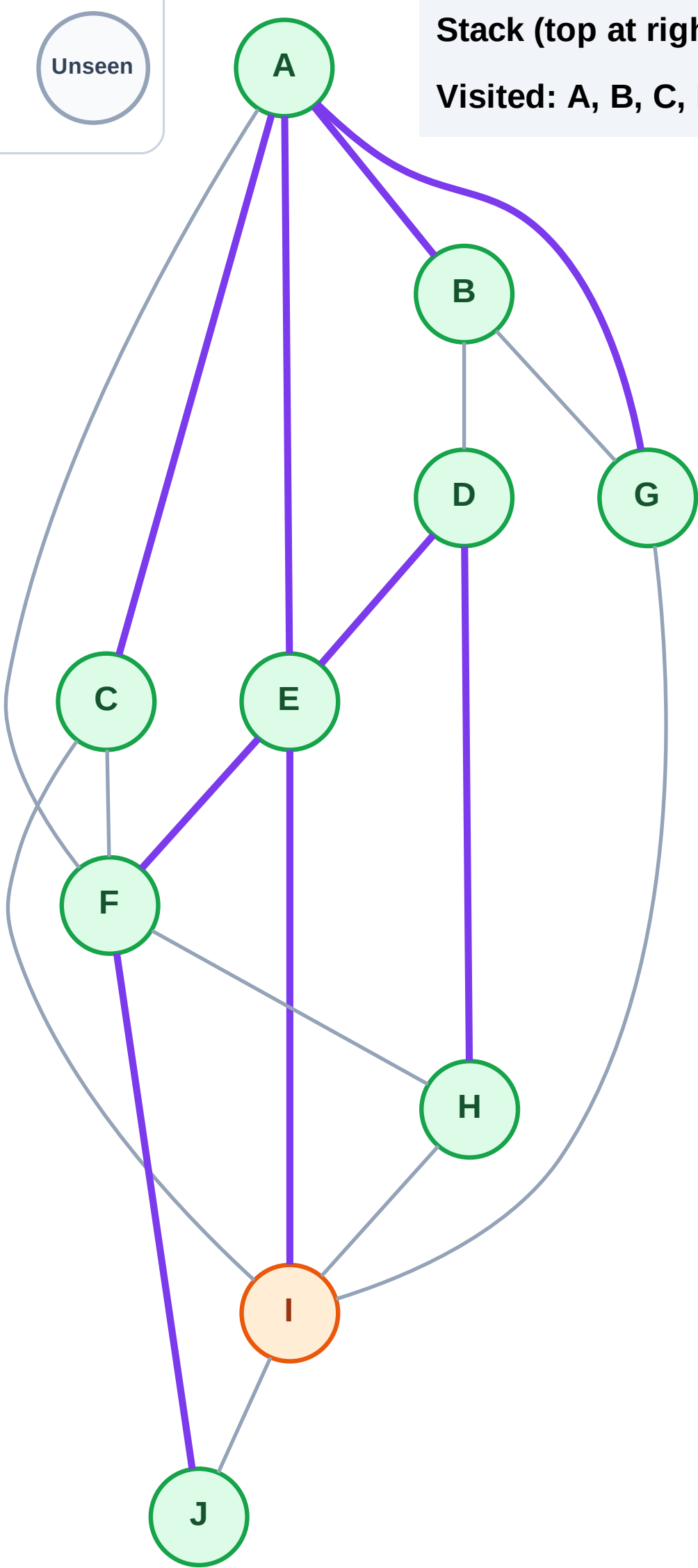
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

